NAVLE Review

Dogs & Cats Study Guide

A complete review of canine and feline medicine, organized by body system.

Toxicology · **Hematology & Hemostasis** · **Cardiology**
Respiratory & Critical Care · **Neurology** · **Ophthalmology**
GI, Liver & Pancreas · **Urinary & Endocrine** · **Dermatology**
Oncology · **Vector-Borne Disease** · **Systemic Mycoses**
Parasitology · **Feline-Specific Medicine** · **Pharmacology & Anesthesia**

Contents

HOW TO USE THIS GUIDE

Yellow boxes are board pearls and buzzword-to-diagnosis shortcuts. **Green boxes** are look-alike discriminators — the "how do I tell these two apart" questions the NAVLE builds whole items around. **Red boxes** flag genuine clinical dangers.

READ APPENDIX C BEFORE YOU PLAN YOUR STUDYING

This guide goes deep on toxicology, clinical pathology, and vector-borne disease. Several guaranteed NAVLE topics — heartworm, the canine and feline endocrinopathies, FIV/FIP, parvovirus, and IVDD — are outside its scope. Appendix C lists them so you can cover them elsewhere.

Z = zoonotic CAT = cat-specific DOG = dog-specific

1 | Toxicology

HOW TOXICOLOGY IS TESTED

Nearly every small animal toxicology item is one of three questions: **What is the mechanism? What is the antidote? What is the one lab abnormality that gives it away?** Build your recall around those three columns and you will recognize almost any stem.

1.1 Rodenticides — Know All Three

Rodenticide	Mechanism	Signs & Lab Findings	Antidote / Treatment
Warfarin *(1st generation anticoagulant)*	Anticoagulant — **interferes with vitamin K–dependent factors II, VII, IX, and X**.	**PROLONGED PT FIRST** — because **factor VII has the shortest half-life** and sits in the **extrinsic** pathway. Hemorrhage, anemia.	**Vitamin K_1 (phytonadione).**
Brodifacoum *(2nd generation — d-CON)*	Inhibits **vitamin K epoxide reductase**. Much longer-acting than warfarin.	**Hemorrhage and anemia.** Prolonged PT then PTT.	**Vitamin K_1** — but for a much longer course (3–4 weeks) because of the extended half-life.
Cholecalciferol	Converted to **active vitamin D** → increased calcium resorption from bone and absorption from gut.	**Fatal HYPERCALCEMIA**, hyperphosphatemia, soft-tissue mineralization, acute kidney injury.	Fluids, furosemide, glucocorticoids, bisphosphonates, calcitonin. **Vitamin K does nothing here.**
Bromethalin *(not vitamin-K responsive)*	**Neurotoxic** — uncouples oxidative phosphorylation in CNS → cerebral edema.	**Tremors, ataxia, paresis.** No coagulopathy.	Decontamination, mannitol for cerebral edema. **No antidote.**

BROMETHALIN VS. BRODIFACOUM — THE CLASSIC PAIRING

Bromethalin = BRAIN. Tremors, ataxia, paresis. Neurotoxic. No vitamin K response.
Brodifacoum = BLEEDING. Hemorrhage and anemia. Responds to vitamin K_1.
If the stem gives you a rodenticide exposure with **neurologic** signs and a **normal** PT, it is not an anticoagulant.

1.2 Ethylene Glycol (Antifreeze)

Finding	Detail
The five classic abnormalities	**1. High anion gap metabolic acidosis** · **2. Azotemia** · **3. Hyperglycemia** · **4. Polyuria progressing to oliguria** · **5. Hypocalcemia** (from chelation by oxalate).
Urine sediment	**Calcium oxalate crystalluria** — the monohydrate form, classically described as "picket fence" or dumbbell-shaped.
Treatment	**4-methylpyrazole (fomepizole, 4-MP)** — competitively inhibits **alcohol dehydrogenase**, preventing conversion of ethylene glycol to its toxic metabolites.
Why 4-MP over ethanol	Three reasons: it does **not** cause **hyperosmolality**, **metabolic acidosis**, or **CNS depression** — all of which ethanol adds on top of the toxicosis.

THE FELINE TREATMENT WINDOW IS RAZOR THIN

Fomepizole IS used in cats — but at a much higher dose than in dogs, and it must be started **within about 3 hours** of ingestion to work. (The older teaching that fomepizole is contraindicated in cats is outdated.) Cats have a far narrower window and a much worse prognosis than dogs, so ethylene glycol exposure in a cat is a true emergency.

1.3 Insecticides & Neurotoxins

Toxin	Mechanism	Signs	Treatment
Organophosphates / carbamates	**Inhibit acetylcholinesterase** → ACh accumulates → muscarinic overstimulation.	**Muscarinic signs: hypersalivation, incoordination, bloat.** Think **SLUDGE** — Salivation, Lacrimation, Urination, Defecation, GI upset, Emesis.	**Atropine** (blocks muscarinic receptors) **and 2-PAM** (pralidoxime) — **2-PAM reactivates acetylcholinesterase** before the enzyme "ages."
Pyrethrins / pyrethroids CAT	**Toxic to CATS.** Alter sodium channel activity to **increase the length of depolarization**. Cats lack the glucuronidation capacity to clear them.	Four outcomes: **depression, hypersalivation, ataxia, muscle tremors.** Classic history: owner applied a dog flea product to the cat.	**Methocarbamol** (muscle relaxant) **and remove the source** — bathe the patient.
Strychnine	Used as **snail/gopher bait**. **Outcompetes glycine**, the inhibitory neurotransmitter of the **spinal cord and brainstem** → loss of inhibition.	Affects **striated (skeletal) muscle** → rigid extensor spasm, "sawhorse" stance, tetanic seizures triggered by stimuli.	**Methocarbamol.** Minimize sensory stimulation; dark, quiet room.
Metaldehyde (snail bait) DOG	The other snail bait. Decreases GABA and serotonin.	**Severe muscle tremors** — "shake and bake." Hyperthermia.	Methocarbamol, cooling, decontamination.
Black widow spider *Latrodectus mactans / L. hesperus*	**Alpha-latrotoxin** triggers massive **acetylcholine** release at the neuromuscular junction, ultimately destroying peripheral nerve terminals.	**Rigid muscle spasm**; progresses to **ascending motor paralysis** and cardiovascular collapse.	Antivenin, muscle relaxants, supportive care.

1.4 Food & Household Toxicoses

Toxin	Mechanism	Signs	Treatment
Xylitol DOG	**Rapid, massive insulin release** from the canine pancreas (does not occur in cats or humans). High doses also cause acute hepatic necrosis.	Three outcomes: **hypoglycemia, hypokalemia, coma and seizures.**	Dextrose IV, potassium supplementation, monitor liver enzymes for 72 hours.
Chocolate	Three toxicants: **methylxanthines** — specifically **theobromine** and **caffeine**. Phosphodiesterase inhibition and adenosine receptor antagonism.	Three outcomes: **CNS excitation, tachycardia, vasoconstriction.** Also vomiting, PU/PD, arrhythmias.	**Highest toxin concentration: unsweetened BAKING chocolate.** (White chocolate is nearly harmless.) Decontamination, fluids, beta-blockers for tachyarrhythmia.
Grapes and raisins	Mechanism now attributed to **tartaric acid**.	**Acute kidney injury.**	Aggressive decontamination and fluid diuresis.
Onions / garlic	Organosulfur compounds cause **oxidative damage** to hemoglobin.	**Heinz body anemia**, plus **hemoglobinuria** and **hemoglobinemia**.	Supportive; transfusion if severe.
Acetaminophen CAT	**Cats lack glucuronyl transferase** and cannot conjugate the drug → toxic metabolites accumulate.	**Methemoglobinemia** — dark brown, "muddy" blood and mucous membranes from oxidative damage to hemoglobin. Also **Heinz body anemia**, facial/paw edema, hepatic necrosis (more so in dogs).	**N-acetylcysteine** (replenishes glutathione). Ascorbic acid as adjunct.

Toxin	Mechanism	Signs	Treatment
Bleach	Corrosive/irritant.	Oral and esophageal irritation, vomiting.	**Dilute the stomach contents** — milk or water. **Do NOT induce emesis** (see correction below).
Fleet (phosphate) enemas CAT	High in **PHOSPHORUS**. Absorbed phosphate chelates calcium.	Three consequences: **decreased serum calcium** → weakness, shock, tremors, seizures; **hypernatremia**; **hyperphosphatemia**.	**Calcium gluconate**, **phosphate binders**, and sometimes **insulin + dextrose** to drive phosphate intracellularly.
Petroleum distillates CAT	Aspiration pneumonitis risk; dermal irritation.	Respiratory distress, dermatitis, pain.	**Buprenorphine** for analgesia. Bathe with mild detergent. **Never induce emesis** — aspiration risk.
Zinc (pennies)	Oxidative hemolysis. **Pennies minted after 1983** are predominantly zinc.	**Heinz body anemia**, hemolysis, GI signs, renal injury.	Remove the foreign body — the anemia will not resolve until the source is out.

NEVER INDUCE EMESIS FOR CORROSIVES OR PETROLEUM DISTILLATES

Emesis is contraindicated for corrosives — bleach, drain cleaners, and strong acids or alkalis — because bringing the corrosive back up re-injures the esophagus and adds aspiration risk. The same rule applies to **petroleum distillates**, where the danger is aspiration pneumonitis.

Correct approach: dilute with milk or water and do NOT induce vomiting.

1.5 Heinz Body Anemia — Memorize the Whole List

The seven general causes	Cat-specific causes
1. **Zinc** 2. **Onions** (and garlic) 3. **Brassica** (kale, cabbage, rape) 4. **Rye grass** 5. **Red maple** (classically the horse) 6. **Molybdenum** 7. **Methylene blue**	1. **Hepatic lipidosis** 2. **Acetaminophen toxicity** 3. **Zinc toxicity** 4. **Diabetic ketoacidosis** *Note that two of these four are metabolic diseases, not poisons — cats form Heinz bodies far more readily than other species because feline hemoglobin has eight reactive sulfhydryl groups.*

RED MAPLE IS A HORSE TOXICITY

Red maple appears on the general Heinz body list, but the classic exam presentation is a **horse** with **icterus**, methemoglobinemia, and hemolytic anemia after eating **wilted** red maple leaves. Keep the species attached to the toxin — molybdenum is likewise a ruminant problem.

1.6 The Lilies — a Guaranteed Question

Lily	Toxic Effect	Detail
True lilies — *Lilium* spp. (Stargazer, Easter, Tiger, Asiatic) and *Hemerocallis* (daylily)	**ACUTE RENAL FAILURE** CAT	The **one of the most toxic lilies for cats** — the **Stargazer / *Lilium* family**. **Every part** is toxic, including pollen and vase water. Cats only; dogs get GI upset. Aggressive fluid diuresis for 48–72 hours.
Peace lily and **Calla lily**	**Insoluble calcium oxalate crystals**	Not true lilies. Cause immediate **oral pain, drooling, and swelling** from mechanical crystal injury — **not** renal failure.

Lily	Toxic Effect	Detail
Lily of the valley *Convallaria majalis*	**CARDIOTOXIC**	Contains **cardiac glycosides** — behaves like digoxin toxicity: arrhythmias, bradycardia, GI signs, hyperkalemia.

THE THREE RENAL TOXICANTS

Lilies · Grapes · Raisins. If a question gives you acute kidney injury in a dog or cat with a dietary or plant history, it is one of these three (or ethylene glycol).

RAPID-FIRE TOXICOLOGY RECALL

Prolonged PT first → **factor VII, shortest half-life, extrinsic pathway** → anticoagulant rodenticide.
Hypercalcemia after rodenticide → **cholecalciferol**.
Tremors + ataxia after rodenticide, normal PT → **bromethalin**.
Calcium oxalate crystalluria + high anion gap acidosis → **ethylene glycol** → **fomepizole**.
SLUDGE → **organophosphate** → **atropine + 2-PAM**.
Cat treated with a dog flea product → **pyrethroid** → **methocarbamol**.
Rigid extensor spasm, snail bait → **strychnine** (glycine antagonist) → **methocarbamol**.
Severe tremors, snail bait → **metaldehyde**.
Dog + sugar-free gum → **xylitol** → hypoglycemia, hypokalemia, seizures.
Brown blood in a cat → **acetaminophen** → **N-acetylcysteine**.
Cat + houseplant + anuria → **true lily**.

2 | Hematology, Hemostasis & Transfusion Medicine

2.1 Coagulation Testing — Which Test, Which Pathway

Test	What It Measures	Clinical Use
PT (prothrombin time)	**Extrinsic and common** pathways	Prolongs **FIRST** in anticoagulant rodenticide toxicosis, because **factor VII** (extrinsic) has the **shortest half-life**.
PTT / aPTT	**Intrinsic and common** pathways	**The test for hemophilia in a dog** — hemophilia A is a factor VIII (intrinsic) deficiency, so you test the intrinsic pathway with aPTT.
Thrombin time	The **final step** of coagulation — **fibrinogen → fibrin**	Detects fibrinogen deficiency or dysfibrinogenemia.
Buccal mucosal bleeding time (BMBT)	**PLATELET FUNCTION**	Normal platelet count with prolonged BMBT → a **qualitative** platelet defect (von Willebrand disease, thrombopathia, NSAIDs).
Template bleeding time	Functional ability of platelets to **plug a minute wound**	Same principle, standardized incision.
Antithrombin (AT III) activity	Anticoagulant reserve	Measured **when DIC is a concern**; also lost in protein-losing nephropathy.

> **FACTOR VII MEMORY HOOK**
>
> **PET** — **P**T tests the **E**xtrinsic pathway, which contains factor Se**VE**n. Vitamin K antagonism hits factor VII first because it dies first.

2.2 Inherited Bleeding Disorders

Disorder	Defect	Abnormal Tests	Treatment / Breed
von Willebrand disease	Deficiency of von Willebrand factor → platelets are **unable to adhere to subendothelial collagen**. The most common inherited bleeding disorder of dogs.	**Prolonged BMBT** with a **normal platelet count**. Doberman is the classic breed.	Cryoprecipitate or fresh frozen plasma; DDAVP.
Hemophilia A	**Inherited deficiency of factor VIII.** X-linked recessive — affected males, carrier females.	**Prolonged ACT and aPTT.** PT is normal.	**Give plasma** (or cryoprecipitate).
Canine thrombopathia	Inherited condition in which **platelets fail to aggregate and to secrete their granules**.	Prolonged BMBT, normal count.	**Basset Hounds** are the breed to know.

2.3 Immune-Mediated Cytopenias

Disease	Diagnostic Findings	Treatment
IMHA (immune-mediated hemolytic anemia)	**Four findings:** 1. **Regenerative anemia** 2. **Spherocytes** on blood smear (plus polychromasia) 3. **Autoagglutination** 4. **Positive Coombs test**	Immunosuppression — glucocorticoids first line, with **azathioprine** as an adjunct (dogs). Antithrombotics, because IMHA is highly thrombogenic.
ITP / IMTP (immune-mediated thrombocytopenia)	Severe thrombocytopenia; petechiae and mucosal bleeding. Diagnosis of exclusion.	Glucocorticoids ± **azathioprine**; vincristine to accelerate platelet release.

AZATHIOPRINE IS CONTRAINDICATED IN CATS

Cats **lack thiopurine methyltransferase (TPMT)** and develop severe, sometimes fatal **bone marrow suppression** on azathioprine. Use **chlorambucil** or **cyclosporine** in cats instead.

SPHEROCYTES BELONG TO IMHA, NOT ITP

Spherocytes — small, dense, hyperchromic RBCs lacking central pallor — indicate **partial phagocytic removal of antibody-coated red cell membrane**. That is an **IMHA** phenomenon.
ITP is diagnosed by a markedly low PLATELET COUNT, not by spherocytes.

2.4 Anemia Classification & Regeneration

Finding	Interpretation
Polychromasia	Sign of a **REGENERATIVE anemia** — less mature RBCs (reticulocytes) being released into circulation.
Microcytic, hypochromic, non-regenerative	**IRON DEFICIENCY.** In an adult dog this means chronic blood loss until proven otherwise — think GI bleeding, or heavy flea/hookworm burden in a young animal.
Mild, non-regenerative, normocytic, normochromic CAT	**Anemia of chronic (inflammatory) disease** — driven by **inflammatory cytokines** sequestering iron. The most common anemia in a sick cat.
Anisocytosis vs. poikilocytosis	**Aniso = variation in SIZE. Poikilo = variation in SHAPE. Poikilocytosis is NORMAL in goats** (and in young cattle) — do not over-read it there.
Stress leukogram	The corticosteroid pattern: **neutrophilia, lymphopenia, monocytosis** (and eosinopenia). Distinguish from an inflammatory leukogram, which has a left shift.

2.5 Congenital Leukocyte Anomalies

Anomaly	Description	Species / Breed
Chédiak-Higashi syndrome	**Large eosinophilic cytoplasmic inclusions in leukocytes** from abnormal granule formation. Causes partial oculocutaneous albinism and a bleeding tendency from defective platelet dense granules.	**Color-dilute (smoke blue) Persian cats.** Also Hereford cattle, Aleutian mink.
Pelger-Huët anomaly	An **apparent severe left shift** — neutrophils have **hyposegmented, band-shaped nuclei** — but with **mature, coarsely clumped chromatin** and **no clinical illness**. A benign inherited defect of nuclear segmentation.	**Australian Shepherds.** The trap: it looks like overwhelming sepsis on a smear from a healthy dog.

2.6 DIC and SIRS

Syndrome	Criteria
DIC *need 2 of 4 to diagnose*	1. **Elevated PT and PTT** 2. **Thrombocytopenia** 3. **Positive D-dimer** (fibrin degradation product) 4. **Decreased antithrombin III** Also expect schistocytes on the smear.
SIRS	Three signs: **fever, tachycardia, leukopenia** (or leukocytosis). Also tachypnea. The cytokines most associated with initiating SIRS are **IL-1** and **TNF-α**.

2.7 Transfusion Medicine

Concept	Detail
Transfusion trigger	**Acute blood loss with PCV < 20%.** (Chronic anemia is tolerated to much lower PCVs — the rate of change matters more than the number.)
Dosing rule	**1 mL/kg of packed RBCs raises the PCV by 1%.**
Worked example	Whisker, a **4 kg** cat, has a PCV of **12%** and you want **25%**. `Target rise = 25% − 12% = 13% Volume = 4 kg × 1 mL/kg × 13 = 52 mL of pRBCs`
Crossmatch — major vs. minor	**MAJOR:** donor **RBCs** + recipient **plasma** — the important one, because it detects recipient antibodies that would destroy the transfused cells. **MINOR:** donor **plasma** + recipient **RBCs**.
Feline blood types CAT	**Type B cats are predisposed to severe transfusion reactions**, because they have strong, naturally occurring anti-A alloantibodies. **Cats must always be typed before a first transfusion** — unlike dogs, they have preformed antibodies with no prior sensitization. **Breeds with type B blood:** Abyssinian, British Shorthair, Devon Rex, Cornish Rex, Himalayan, Persian, Somali, and Scottish Fold.
Bovine blood types	The two clinically relevant systems are **B and J**.
Treating a transfusion reaction	**Stop the transfusion**, then: **dexamethasone/steroid, epinephrine, antihistamine, and oxygen.**

NEVER GIVE TYPE A BLOOD TO A TYPE B CAT

Type B cats have very strong, naturally occurring anti-A antibodies — giving type A blood to a type B cat causes an acute, often fatal hemolytic reaction, sometimes within milliliters. Type A cats have only weak anti-B antibodies, so a mismatch in that direction is less immediately catastrophic, though still wrong.
The rule: always blood-type cats before any transfusion. Cats have preformed antibodies and need no prior sensitization to react.

2.8 Calcium Disorders

Topic	Detail
First step in evaluating hypocalcemia	**Check for concurrent hypoproteinemia / hypoalbuminemia first**, then measure **ionized** calcium. Roughly half of total calcium is protein-bound, so a low albumin gives a falsely low total calcium with a normal ionized fraction and no clinical signs.
Causes of hypocalcemia	**Renal disease · Eclampsia** (periparturient tetany in a nursing bitch) **· Phosphate enema toxicity · Ethylene glycol toxicosis · Hypoparathyroidism.** **The three to name first in a dog or cat: eclampsia, hypoparathyroidism, and ethylene glycol toxicity.**
Causes of hypercalcemia — "HARD IONS"	**H**yperparathyroidism · **A**ddison's (hypoadrenocorticism) · **R**enal disease/CKD · **D** — vitamin D toxicosis **I**diopathic (classically the cat) · **O**steolytic · **N**eoplastic · **S**purious *The two you should think of first in a dog:* ***lymphoma*** *and* ***anal sac apocrine gland adenocarcinoma*** *— both hypercalcemia of malignancy.*

2.9 Joint Fluid Analysis

Classification	Cytology
Normal	**Small mononuclear cells without neutrophils.**
Suppurative	**Neutrophils.** Think septic arthritis or immune-mediated polyarthritis.

Classification	Cytology
Granulomatous	**Mononuclear cells** — lymphocytes, macrophages, plasma cells.
Pyogranulomatous	**Mixed** neutrophils and mononuclear cells.

2.10 Hypersensitivity Reactions

Type	Mechanism	Examples
Type I	**IgE-mediated**, immediate	Allergy, atopy, anaphylaxis, vaccine reactions.
Type II	**Antibody-dependent** — IgG or IgM made against normal self antigen	IMHA, ITP, myasthenia gravis, neonatal isoerythrolysis.
Type III	**Antigen–antibody (immune) complex** formation overwhelms clearance and deposits in tissue	Causes **arthritis, nephritis (glomerulonephritis), and uveitis** — the classic triad of deposition sites. **Sulfa drugs cause a type III reaction in Doberman Pinschers.** Also: purpura hemorrhagica, Aleutian disease of ferrets/mink.
Type IV	**Delayed, cell-mediated** — T lymphocytes become sensitized and differentiate into cytotoxic T cells	Five to name: **contact dermatitis**; **granulomatous reactions** (mycobacteria, *Coccidioides*, *Blastomyces*, *Histoplasma*); **lymphocytic choriomeningitis** in hamsters and rats; **canine distemper**; **autoimmune thyroiditis**; and **KCS**. Also transplant rejection and tumor immunity.

RAPID-FIRE HEMATOLOGY RECALL

Spherocytes + autoagglutination + positive Coombs → **IMHA**.
Normal platelet count but prolonged BMBT → **von Willebrand disease**.
Prolonged aPTT in a young male dog → **hemophilia A** → give **plasma**.
Basset Hound that bleeds with normal counts → **canine thrombopathia**.
Blue Persian with giant leukocyte granules → **Chédiak-Higashi**.
Australian Shepherd with a "severe left shift" but perfectly healthy → **Pelger-Huët**.
Microcytic hypochromic → **iron deficiency = chronic blood loss**.
1 mL/kg pRBCs → **1% PCV**.
Never give type A blood to a **type B cat**.

3 | Cardiology

3.1 Hemodynamic First Principles

Concept	Definition
Cardiac output	**CO = Stroke Volume × Heart Rate**
Stroke volume determined by	**Preload · Afterload · Contractility**
Oxygen delivery	**DO_2 = Cardiac Output × Arterial Oxygen Content**
Compensation for acute blood loss	Three responses: **1.** Increased cardiac output with vasoconstriction (RAAS/angiotensin). **2.** ADH drives water and sodium resorption by the kidney. **3. Splenic contraction** autotransfuses stored red cells.

Body water compartments

Compartment	Fraction
Total body water (TBW)	**~60% of body weight.** Use **70%** for pediatric patients and **lean** body weight for obese patients (fat holds little water).
Intracellular fluid (ICF)	**2/3 of TBW = 40% of body weight**
Extracellular fluid (ECF)	**1/3 of TBW = 20% of body weight**
ECF subdivides into	**Interstitial = 75% of ECF** · **Intravascular = 25% of ECF** *This is why only about a quarter of a crystalloid bolus stays in the vasculature.*

3.2 Shock

Type	Mechanism
Hypovolemic	Critical reduction in **intravascular volume** — hemorrhage, third-spacing, or dehydration → decreased tissue perfusion and oxygenation.
Cardiogenic	**Decreased myocardial contractility** with subsequent fall in oxygen delivery. **Always associated with primary heart disease.**
Obstructive	Abnormal blood distribution that **impairs venous return** — **GDV** compressing the caudal vena cava, **pericardial tamponade**, or saddle thrombus.
Distributive (vasogenic)	Typically secondary to **sepsis or anaphylaxis**, causing widespread **vasodilation** and maldistribution of an adequate blood volume.

3.3 Heart Sounds & Murmur Localization

Sound	Origin
S1	**Closure of the AV valves** (mitral and tricuspid) — marks the onset of ventricular contraction, **systole**.
S2	**Closure of the pulmonic and aortic valves** — onset of **diastole**.
S3	**Rapid ventricular filling / relaxation.** A pathologic gallop in dogs and cats — think dilated cardiomyopathy.
S4	**Atrial contraction.** Pathologic gallop — think hypertrophic cardiomyopathy in cats.

LEFT-SIDE AUSCULTATION — "PAM"

Pulmonic — 3rd intercostal space
Aortic — 4th intercostal space
Mitral — 5th intercostal space
On the **right**, the **tricuspid** valve sits between the **3rd and 4th ICS**. **Tricuspid dysplasia** produces a **systolic murmur at the right mid-thorax**.

Location	Murmurs Heard There
Left heart BASE	Three: **pulmonic stenosis**, **PDA**, and **subaortic stenosis**.
Left heart APEX	**Mitral valve dysplasia** (and acquired mitral regurgitation — by far the most common murmur in an older small-breed dog).
Right side, 3rd–4th ICS	**Tricuspid** disease; also where a **VSD** is often loudest.

3.4 Congenital Heart Disease

Defect	Murmur	Key Features	Treatment
PDA DOG	**Continuous "machinery" or "washing machine" murmur**, loudest at the left heart base.	**The most common congenital defect in DOGS.** Bounding, hyperkinetic femoral pulses.	**Surgical ligation or coil embolization.** **Do NOT close a right-to-left (reversed) shunt** — the shunt is the patient's only pressure relief, and closing it is fatal.
VSD CAT	**Loud holosystolic (constant) murmur heard on BOTH sides.**	**The most common congenital defect in CATS**, and also **the most common congenital defect in CATTLE** (where it presents at 2–3 months).	Small defects often tolerated; large ones lead to CHF.
Patent foramen ovale / ASD	Often soft or a relative pulmonic stenosis murmur.	Fails to close within **48 hours of birth** → **shunting from the LEFT atrium to the RIGHT atrium**.	Usually incidental unless large.
Tetralogy of Fallot	Variable; often a pulmonic stenosis murmur.	**Four components:** 1. **Overriding (dextroposed) aorta** 2. **Pulmonic stenosis** 3. **Ventricular septal defect** 4. **Right ventricular hypertrophy** **Unique finding: POLYCYTHEMIA.**	The **right-to-left shunt** causes hypoxemia → increased **erythropoietin** → polycythemia and **dark red mucous membranes**. Phlebotomy; palliative shunting.

3.5 Acquired Disease

Condition	Key Facts
Valvular endocarditis — which valve?	**Small animals: the AORTIC valve** (and mitral). **Cattle: the TRICUSPID valve / right side.** The most commonly isolated organism from vegetative endocarditis is ***Streptococcus***; in dogs also consider ***Bartonella vinsonii*** — a dog with a new murmur and no flea/tick preventive.

Condition	Key Facts
Pericardial effusion	**Five signs:** 1. **Muffled heart sounds** 2. **Electrical alternans** — beat-to-beat variation in QRS amplitude as the heart swings in fluid 3. **Pulsus paradoxus** — systolic blood pressure drops more than 10 mmHg on inspiration 4. **Enlarged, globoid heart** on the DV/VD radiograph 5. **"Nutmeg" liver** on necropsy (chronic passive congestion) Also jugular distension, ascites, weak pulses. **Treatment: pericardiocentesis.**

3.6 Arrhythmias

Arrhythmia	Recognition	Treatment
Atrial fibrillation — small animals	**Irregularly irregular rhythm with variable heart sound intensity**; no P waves; pulse deficits.	**Four drugs: diltiazem** (calcium channel blocker — decreases heart rate), **atenolol** (beta blocker — rate and rhythm), **digoxin** (rate control), and **procainamide**.
Atrial fibrillation — cattle	**The most common arrhythmia in cattle.** Sign: **pulsus alternans** — two quick normal pulses in a row followed by a dropped pulse.	Often resolves with correction of the underlying GI disease.
Atrial fibrillation — horses	**Quinidine** or another **class IA sodium channel blocker**. Add **digoxin** for rate control if quinidine produces an accelerated ventricular response.	Contrast: the most common **pathologic** arrhythmia in horses is **2nd degree AV block** (though physiologic 2° AVB at rest is normal in horses).
Atrial premature contractions	Diagnosed clinically by **pulse deficits**.	Treat the underlying disease.
VPCs — when to treat	Four indications: 1. **HR > 180 bpm** 2. **Pulse deficits** 3. **Clinical signs** (syncope, weakness) 4. **VPCs / ventricular tachycardia lasting > 20 seconds** Also R-on-T phenomenon and multiform complexes.	Lidocaine acutely; **mexiletine** or sotalol for long-term management.
Ventricular fibrillation	**Chaotic electrical AND mechanical activity** during cardiac arrest.	Defibrillation.
Ventricular asystole	**NO electrical and no mechanical activity.** A flat line.	Not shockable — CPR and epinephrine.

ECG IN HYPERKALEMIA — THE PROGRESSION

1. Tented (tall, spiked) T waves — earliest change
2. Prolonged P-R interval
3. Loss of the P wave (atrial standstill)
4. Widened QRS → sine wave → arrest
Classic setting: **blocked male cat**, uroabdomen, or hypoadrenocorticism. Treat with calcium gluconate (cardioprotection), then insulin+dextrose or a beta agonist to shift potassium intracellularly.

Atrial enlargement on ECG

Chamber	P Wave Change
Right atrial enlargement (P pulmonale)	**Tall and slender P waves** — increased amplitude.
Left atrial enlargement (P mitrale)	**Increased P wave DURATION** — a wide, often notched P.

3.7 Heart Failure Therapy

Drug	Role
Furosemide	**Loop diuretic** — the cornerstone of acute CHF management. **In a cat in CHF, the first drug to give is IV or IM furosemide.**
Nitroprusside	**Venodilator** (and arteriodilator) for fulminant, refractory pulmonary edema. Requires blood pressure monitoring.
Enalapril / benazepril	**ACE inhibitors** — block angiotensin II formation, stopping vasoconstriction and reducing aldosterone. **Enalapril increases cardiac output by reducing afterload.**
Spironolactone	**Aldosterone antagonist**, potassium-sparing diuretic. **Also a competitive antagonist at the ADH/ aldosterone axis.**
Sodium-restricted diet	Adjunct to reduce volume retention.
Diltiazem	**Calcium channel blocker** treating three things: **atrial fibrillation, supraventricular tachycardia, and HCM.**
Amlodipine	**Calcium channel blocker for HYPERTENSION** — the first-line antihypertensive in cats, and therefore the drug that prevents hypertensive **posterior uveitis and retinal detachment**.
Horse in CHF	**Furosemide and benazepril.**

PIMOBENDAN

Pimobendan is a first-line drug for canine CHF from myxomatous mitral valve disease and DCM. It is an **INODILATOR** — a calcium sensitizer that increases contractility, plus a phosphodiesterase-3 inhibitor that vasodilates. It is also used pre-clinically in stage B2 mitral valve disease to delay the onset of heart failure.

RAPID-FIRE CARDIOLOGY RECALL

Continuous machinery murmur → **PDA** → most common congenital defect in **dogs**.
Holosystolic murmur on both sides → **VSD** → most common in **cats** and **cattle**.
Polycythemia + dark red mucous membranes + congenital murmur → **Tetralogy of Fallot**.
Muffled sounds + electrical alternans + globoid heart → **pericardial effusion** → **pericardiocentesis**.
Tented T waves, no P waves → **hyperkalemia**.
Irregularly irregular with pulse deficits → **atrial fibrillation**.
Endocarditis: **aortic** valve in small animals, **tricuspid** in cattle.
Cat in CHF → **furosemide first**.
Hypertension in a cat → **amlodipine**.

4 | Respiratory, Acid–Base & Critical Care

4.1 Oxygenation & Ventilation

Concept	Detail
Ventilation is defined by $PaCO_2$	**Normal: 35–45 mmHg** **Hyperventilating: $PaCO_2$ < 30 mmHg** **Hypoventilating: $PaCO_2$ > 45 mmHg** *Ventilation is about CO_2, not oxygen. This is the single most useful acid–base fact.*
Cyanosis	Indicates **PaO_2 less than 50 mmHg**. It requires ≥5 g/dL of desaturated hemoglobin — which is why **a severely anemic patient can be profoundly hypoxemic and never look cyanotic**.
Five causes of hypoxemia	1. **Hypoventilation** · 2. **Low FiO_2** · 3. **V/Q mismatch** · 4. **Diffusion impairment** · 5. **Venous admixture / right-to-left shunt**
Most common cause of cardiac arrest	**Systemic hypoxemia.** This is why airway and breathing come before everything else.

Pulse oximetry → arterial oxygen conversion

SpO_2	PaO_2	Interpretation
98–100%	**> 100 mmHg**	Normal on room air or supplemented.
95%	**80 mmHg**	Borderline — start paying attention.
90%	**60 mmHg**	**The clinical cutoff.** Below this you are on the steep part of the curve and desaturating fast. **Supplement oxygen.**
50%	**30 mmHg**	Critical.
10%	**10 mmHg**	Incompatible with life.

THE 90/60 RULE

An SpO_2 of **90% corresponds to a PaO_2 of 60 mmHg** — memorize this one pair and you can reconstruct the rest of the oxyhemoglobin dissociation curve. Below 90%, small further drops in saturation mean large drops in PaO_2.

4.2 Acid–Base

Disturbance	Causes & Species Notes
Metabolic acidosis in cattle	**Saliva loss** — ruminant saliva is **rich in bicarbonate**, so losing it (esophageal obstruction, salivation) costs the animal its buffer. Also grain overload/rumen acidosis and diarrhea.
Metabolic alkalosis in horses	**Saliva loss** in the horse causes **alkalosis** instead, because equine saliva is **rich in chloride**. Also the **sweating horse**, which loses large amounts of chloride and potassium.
Respiratory acidosis	**Hypoventilation** with buildup of CO_2 — airway obstruction, pneumothorax, flail chest, pleural space disease, anesthetic depression.
Normal HCO_3^-	**22–24 mEq/L**
Base excess	The **amount of base needed to return plasma pH to normal**. A negative value (base deficit) means metabolic acidosis.

Bicarbonate deficit calculation

Formula:

```
HCO₃⁻ needed (mEq) = 0.3 to 0.4 × body weight (kg) × base deficit
```

Worked example — a **470 kg** horse with a base excess of **−13**, correcting fully:

```
0.4 × 470 kg × 13 = ~2,440 mEq of sodium bicarbonate
```

In practice you would give roughly half this and recheck. The 0.3 factor approximates the ECF volume; 0.4 is used when a more complete correction is intended.

4.3 Pleural Space & Pulmonary Disease

Condition	Key Feature	Management
Pneumothorax	**NO lung sounds DORSALLY** — free air rises to the highest point in the chest.	**Thoracocentesis at the DORSAL aspect** of the chest, where the air is. (Fluid is tapped ventrally.)
Aspiration pneumonia	**Alveolar pattern in the RIGHT CRANIAL and RIGHT MIDDLE lung lobes** — gravity and bronchial anatomy direct aspirate there in a sternally recumbent animal.	Antibiotics, oxygen, nebulization/coupage. Look for the underlying cause: megaesophagus, laryngeal paralysis, anesthesia, seizures.
Atelectasis	**Incomplete expansion of lung from loss of air in the alveoli.** Three common causes: **prolonged recumbency**, **inhalation anesthesia**, and **decreased surfactant**.	Recruitment maneuvers, positional changes, PEEP.
Neurogenic pulmonary edema	**CAUDODORSAL** distribution on radiographs — the opposite of the perihilar/caudodorsal cardiogenic pattern in dogs, and a useful discriminator. Four causes: **head trauma, seizures, electrocution, upper airway obstruction** (including strangulation and choke).	Oxygen, treat the primary insult. Usually self-limiting over 24–72 hours.
Lung lobe torsion	Typically the right middle lobe in deep-chested dogs, or any lobe following chronic effusion.	**Lung lobectomy.** During surgery **isoflurane plus succinylcholine** (a depolarizing neuromuscular blocker) is used to keep the patient completely still.

OXYGEN IN PNEUMOTHORAX

Oxygen is supportive and not contraindicated — by creating a nitrogen gradient it actually speeds reabsorption of pleural air. But **oxygen alone is not sufficient treatment: you must remove the air.** Do both.

4.4 Trauma & Emergency

Topic	Detail
Smoke inhalation — four effects	1. **Carbon monoxide production** → displaces oxygen from hemoglobin (and pulse oximetry reads falsely normal). 2. **Carbon dioxide production** → severe acidosis. 3. **Laryngeal edema** → upper airway obstruction (may be delayed hours). 4. **Inhibition of pulmonary macrophage function** → secondary bacterial pneumonia days later.
Burn classification	**1st degree** — epidermis only. **2nd degree** — epidermis and deep dermis; **partial thickness**. **3rd degree** — epidermis, dermis, and adnexal structures (full thickness — no regenerative source left). **4th degree** — total destruction of skin, fat, fascia, muscle, and bone.
Septic peritonitis — the diagnostic rule	**Blood glucose more than 20 mg/dL HIGHER than the abdominal fluid glucose** confirms septic peritonitis. Bacteria consume glucose in the effusion. Also: seeing intracellular bacteria in effusion neutrophils is definitive.

Topic	Detail
Uroabdomen — the diagnostic ratios	**Abdominal fluid : serum CREATININE ≥ 2:1** **Abdominal fluid : serum POTASSIUM ≥ 1.4:1 in dogs, 1.9:1 in cats** *Creatinine is the more reliable of the two because it is a larger molecule and equilibrates across the peritoneum more slowly.*
Cushing reflex	**Bradycardia and hypertension in response to increased intracranial pressure.** **Do not lower the blood pressure** — the hypertension is a compensatory response maintaining cerebral perfusion against the rising ICP. Lowering it stops brain blood flow. Treat the ICP instead (mannitol, hypertonic saline, elevate the head 30°).
Lactate	A reflection of **ANAEROBIC metabolism** — produced from **pyruvate** in an anaerobic environment to regenerate NAD^+ and keep glycolysis running. **Normal: < 2.5 mmol/L.** Serial lactate (clearance) is a better prognostic tool than any single value.

RAPID-FIRE CRITICAL CARE RECALL

No lung sounds **dorsally** → **pneumothorax** → tap **dorsally**.
Alveolar pattern right cranial/middle → **aspiration pneumonia**.
Caudodorsal edema after a seizure or choke → **neurogenic pulmonary edema**.
Bradycardia + hypertension in a head-trauma patient → **Cushing reflex — do not treat the blood pressure**.
Abdominal fluid creatinine twice serum → **uroabdomen**.
Abdominal fluid glucose 20 mg/dL below blood → **septic peritonitis**.
$PaCO_2$ > 45 → **hypoventilation**. $PaCO_2$ < 30 → **hyperventilation**.
Normal lactate → **< 2.5 mmol/L**.

5 | Neurology

THE NEUROLOGY EXAM STRATEGY

Localize first, then differentiate. Almost every NAVLE neurology item can be answered by asking two questions: **UMN or LMN?** and **which spinal segment or cranial nerve?** The disease name follows from the localization.

5.1 UMN vs. LMN

Sign	Upper Motor Neuron	Lower Motor Neuron
Reflexes	**Hyper-reflexia** (loss of descending inhibition)	**Decreased or absent** spinal reflexes
Muscle tone	**Increased**	**Decreased** (flaccid)
Muscle atrophy	Slow, disuse atrophy	**Rapid, severe neurogenic atrophy**

5.2 Spinal Cord Localization — The Four Segments

Segment	Forelimbs	Hindlimbs / Other
C1–C5	**UMN**	**UMN** — all four limbs UMN.
C6–T2	**LMN** (the brachial intumescence)	**UMN** in the hindlimbs.
T3–L3	**Normal**	**UMN** in the hindlimbs. *The most common site of IVDD in chondrodystrophic dogs.*
L4–S4	**Normal**	**LMN** in the hindlimbs, **anus, and bladder → incontinence**.

Reflexes and their nerves

Reflex	Spinal Segments	Nerve
Patellar	**L4–L6**	**Femoral nerve**
Withdrawal (pelvic limb)	**L6–S1**	**Sciatic nerve**

Bladder function by lesion location

Lesion	Bladder
Cranial to L7 (UMN bladder)	**Turgid and DIFFICULT to express.** The detrusor contracts against a spastic, non-relaxing urethral sphincter. Risk of bladder rupture and detrusor damage — express or catheterize regularly.
Caudal to L7 (LMN bladder)	**Flaccid and EASY to express, with no anal tone.** Constant overflow dribbling.

5.3 Cranial Nerves

#	Nerve	Function & Lesion Signs
I	**Olfactory**	Smell.
II	**Optic**	Vision — assess with **PLR and dazzle reflex**. Afferent arm of the PLR and menace.
III	**Oculomotor**	Movement of the eye; eyelid elevation; **pupillary constriction** (parasympathetic to the iris). Lesion → mydriasis and ventrolateral strabismus.
IV	**Trochlear**	Innervates the **dorsal oblique**. Lesion → **dorsomedial rotation of the globe** (best seen in species with a horizontal pupil).

#	Nerve	Function & Lesion Signs
V	**Trigeminal**	**Mandibular** — motor to muscles of mastication; damage → **dropped jaw, cannot close the mouth**. **Maxillary** — sensation to face and nose. **Ophthalmic** — sensory to eye, cornea, eyelids (afferent of the palpebral and corneal reflex).
VI	**Abducens**	Controls the **lateral rectus**. Lesion → **MEDIAL strabismus** (the eye cannot move outward).
VII	**Facial**	Facial expression; **closes** the eyelids (orbicularis oculi); taste to the rostral 2/3 of the tongue; lacrimation. Lesion → **loss of menace and palpebral response, ear droop, lip droop**, and **corneal ulceration from exposure keratitis**.
VIII	**Vestibulocochlear**	Balance and hearing. Lesion → **nystagmus**, head tilt, ataxia.
IX	**Glossopharyngeal**	Swallowing (with X); gag reflex.
X	**Vagus**	Lesion → **dysphonia, dysphagia, or megaesophagus**; laryngeal paralysis.
XI	**Accessory**	Motor to trapezius and neck muscles.
XII	**Hypoglossal**	Tongue motor. Lesion → tongue deviation/atrophy.

5.4 Vestibular Disease

Feature	PERIPHERAL	CENTRAL
Nystagmus	**Horizontal or rotary**, direction does not change with head position	**VERTICAL** nystagmus, or a nystagmus that changes direction with head position
Proprioceptive deficits	**ABSENT** — the hallmark	**PRESENT** — the hallmark
Other	Horner's syndrome, facial nerve paralysis (both run through the middle ear)	Altered mentation, other cranial nerve deficits, cerebellar signs

PARADOXICAL VESTIBULAR DISEASE

The head tilt and nystagmus point AWAY from the lesion. Caused by a lesion in the cerebellar peduncle or flocculonodular lobe. **The rule that saves you:** when a vestibular case has both vestibular signs and **CP deficits**, **the lesion is on the side of the PROPRIOCEPTIVE deficits** — proprioception never lies. Trust the CP deficits over the head tilt.
*Classic example: a lesion in the **left cerebellum** producing **left hypermetria** and a **right** paradoxical head tilt — GME is a common cause.*

5.5 Seizures & Brain Disease

Condition	Key Facts
Status epilepticus	**Diazepam — including per rectum**, which works even in cats and is the route to know when there is no IV access. Follow with a loading dose of levetiracetam or phenobarbital.
Idiopathic epilepsy — the signalment	**Age 1–5 years** is the defining feature — a first seizure inside that window in an otherwise normal dog with a normal interictal exam. **Breeds:** Beagle, Keeshond, Dachshund, Labrador Retriever, Golden Retriever, Vizsla. Also Border Collie, German Shepherd, Belgian Tervuren. *A first seizure **under 1 year** suggests congenital/portosystemic shunt or infection; **over 5–6 years** suggests neoplasia or metabolic disease.*
Granulomatous meningoencephalitis (GME)	Idiopathic inflammatory CNS disease of young to middle-aged small-breed dogs. Can produce **paradoxical** vestibular signs.
Syringomyelia	**Development of fluid-filled cavities within the spinal cord.** Classic in Cavalier King Charles Spaniels with Chiari-like malformation — phantom scratching at the shoulder, neck pain.

Condition	Key Facts
Schiff-Sherrington posture	**Rigid forelimb extension with NORMAL forelimb function, plus hindlimb paralysis or paresis.** Caused by **interruption of ascending inhibitory input from border cells in the lumbar spinal cord** to the forelimb extensors. **It indicates a severe T3–L3 lesion — it does NOT indicate a poor prognosis by itself**, and it is not a forelimb lesion. That is the trap.
Myasthenia gravis	**Best diagnostic test: acetylcholine receptor antibody assay** (serum). A type II hypersensitivity. Presents with exercise-induced weakness and, importantly, **megaesophagus** → aspiration pneumonia.
Thiamine (vitamin B_1) deficiency CAT	Seen in **cats fed raw seafood**, which contains **thiaminase** that destroys dietary thiamine. Signs: ventroflexion of the neck, ataxia, mydriasis, and characteristic **dorsal flexion of the head ("stargazing")**, progressing to seizures. Rapidly reversible with thiamine if caught early.

5.6 Horner's Syndrome

Component	Detail
The four signs	1. **Miosis** (constricted pupil) 2. **Ptosis** (drooped upper eyelid) 3. **Enophthalmos** (sunken globe) 4. **Third eyelid protrusion**
Species additions	**Horses:** ipsilateral **head and neck sweating**. **Cattle:** regional **hyperthermia** / loss of sweating on the muzzle.
Localization	Loss of sympathetic innervation anywhere along a three-neuron pathway. The two most commonly tested sites are the **T1–T3 spinal cord** and **otitis media**. Also brachial plexus avulsion and retrobulbar disease.

5.7 Large Animal Neuro Look-Alikes Worth Knowing

Disease	Discriminating Features
Listeriosis ("circling disease")	**UNILATERAL cranial nerve deficits** — classically a **dropped lip/ear** and dysphagia. CSF: **MONONUCLEAR pleocytosis**. Silage-associated. **Treatment: procaine penicillin** (high dose, prolonged).
TEME (thromboembolic meningoencephalitis, *Histophilus somni*)	**Fever and respiratory signs come FIRST**, before the CNS signs. CSF: **NEUTROPHILIC pleocytosis** and **xanthochromic (yellow) fluid** indicating prior hemorrhage. Feedlot cattle.

RAPID-FIRE NEUROLOGY RECALL

Vertical nystagmus or CP deficits → **CENTRAL** vestibular.
Head tilt one way, CP deficits the other → **paradoxical — trust the CP deficits**.
Rigid forelimbs, paralyzed hindlimbs → **Schiff-Sherrington, T3–L3**.
Turgid bladder you can't express → **UMN, lesion cranial to L7**.
Flaccid bladder, no anal tone → **LMN, lesion caudal to L7**.
Medial strabismus → **CN VI abducens**.
Can't blink, corneal ulcer → **CN VII facial**.
Megaesophagus + weakness → **myasthenia gravis** → **ACh receptor antibody assay**.
Cat fed raw fish, ventroflexed neck → **thiamine deficiency**.
Unilateral dropped lip + mononuclear CSF in a cow → **Listeria** → **procaine penicillin**.

6 | Ophthalmology

6.1 Uveitis

Type	Structures Affected	Causes & Consequences
Anterior uveitis	Three structures: **anterior chamber, iris, and ciliary body.**	Aqueous flare, hypopyon, miosis, low IOP, ocular pain. **In cats, chronic anterior uveitis is the usual cause of CATARACTS** — always look for the underlying systemic disease (FeLV, FIV, FIP, toxoplasmosis, lymphoma).
Posterior uveitis	Two structures: **choroid and retina** (chorioretinitis).	**Caused by HYPERTENSION** — which is why the answer to a hypertensive cat with retinal detachment or intraocular hemorrhage is **amlodipine**. Also infectious and neoplastic causes.

6.2 Pharmacology of the Eye

Drug	Class / Mechanism	Use
Pilocarpine	**Muscarinic receptor AGONIST**	**Pupil constrictor.** Used to **diagnose CN III (oculomotor) lesions** — a denervated pupil is supersensitive and constricts to dilute pilocarpine. Also lowers intraocular pressure.
Atropine, glycopyrrolate, tropicamide	**Anticholinergic (parasympatholytic)**	**Mydriasis.** Atropine also relieves ciliary spasm pain in uveitis. Tropicamide is the short-acting diagnostic dilator.
Timolol	**Beta blocker** acting on the ciliary body	**Decreases aqueous humor production** → lowers IOP in glaucoma.
Dorzolamide	**Carbonic anhydrase inhibitor**	**Reduces IOP by decreasing aqueous humor production.**
Latanoprost	**Prostaglandin analogue**	**Increases aqueous humor OUTFLOW** (uveoscleral). Potent IOP reduction — but **contraindicated in uveitis and in feline glaucoma**.

6.3 Miscellaneous Ocular

Condition	Key Fact
Feline herpesvirus (FHV-1) ocular disease CAT	Causes **conjunctivitis and keratitis** — classically **dendritic corneal ulcers**, which are pathognomonic. Treat with **famciclovir**.
Limbal (epibulbar) melanoma	**Slow growing with a low rate of metastasis** — a benign-behaving tumor at the limbus. Contrast with intraocular uveal melanoma in cats, which is far more aggressive.
Posterior synechia	Adhesion of the **iris to the anterior lens capsule**, a sequela of anterior uveitis. Produces an irregular, "dyscoric" pupil.
KCS	A **type IV hypersensitivity** against the lacrimal gland. Also drug-induced — **sulfonamides/TMS** are the classic culprit.
Equine glaucoma vs. uveitis pupil	**Active glaucoma** → **MYDRIATIC pupil. Uveitis without glaucoma** → **MIOTIC pupil.** Treat equine recurrent uveitis glaucoma with **timolol and dorzolamide**.

7 | GI, Liver & Pancreas

7.1 Liver Enzymes & Function

Category	Detail
LEAKAGE enzymes	**ALT and AST** — released from damaged hepatocytes. They indicate **hepatocellular injury** and are **NOT a measure of liver function**.
CHOLESTATIC (inducible) enzymes	**ALP and GGT** — induced by cholestasis (and, in dogs, by corticosteroids and phenobarbital).
Where ALP comes from — four sources	**Liver · Intestine · Kidney · BONE** (the bone isoenzyme is why young growing dogs and dogs with osteosarcoma have high ALP). *The steroid-induced isoenzyme is dog-specific and does not exist in cats — which is why **any** ALP elevation in a cat is significant.*
Liver-specific enzyme in large animals	**SDH (sorbitol dehydrogenase)** in cow, sheep, horse, and goat. ALT is not liver-specific in these species.
Best measure of liver FUNCTION	**Bile acids** (pre- and post-prandial) — or ammonia. See the correction below.
What the liver produces	**Bilirubin · bile acids · fibrinogen · albumin · antithrombin III · BUN (urea) · cholesterol · glucose · coagulation factors.** *Loss of any of these is how liver failure presents: hypoalbuminemia, coagulopathy, low BUN, hypoglycemia, hypocholesterolemia.*

7.2 Portosystemic Shunt (PSS)

Feature	Detail
Pathognomonic in CATS CAT	**Bright copper/orange irises** and **ptyalism (drooling).** The drooling in particular is a much more prominent PSS sign in cats than in dogs.
Associated anemia	**MICROCYTIC anemia** — from abnormal iron transport and sequestration, not iron deficiency per se.
Signalment	Young animal, stunted growth, post-prandial neurologic signs (hepatic encephalopathy), ammonium biurate uroliths.
Three medical treatments	1. **Lactulose** — acidifies the colon, **trapping ammonia in the ionized (NH_4^+) form to decrease absorption**. 2. **Feed the maximum amount of protein tolerated without causing encephalopathy** — restriction should not be so severe that it worsens catabolism. 3. **Oral neomycin or metronidazole** — kills urease-producing gut flora that generate ammonia. *Definitive treatment is surgical attenuation of the shunting vessel.*

7.3 Feline Hepatobiliary & GI Disease

Condition	Key Facts
Feline triaditis CAT	**Extrahepatic biliary obstruction in cats is associated with the triad of:** 1. **Pancreatitis** 2. **IBD** 3. **Cholangitis / cholangiohepatitis** *Anatomic basis: in the cat the pancreatic duct and common bile duct fuse **before** entering the duodenum, so inflammation ascends readily between all three organs.*
Cholangitis treatment CAT	**Amoxicillin-clavulanic acid** is the antibiotic of choice — good coverage of the ascending enteric organisms and excellent biliary penetration.

Condition	Key Facts
Feline colitis — three common causes	1. ***Tritrichomonas foetus*** 2. **Adverse food reaction** 3. **Lymphoplasmacytic colitis**
Ollulanus tricuspis CAT	**The stomach worm of CATS ONLY.** Causes **chronic vomiting** and gastritis. Diagnosed by examining vomitus, not feces — the larvae are passed in vomit.
Platynosomum fastosum CAT	**The liver fluke of cats** causing **hepatobiliary AND pancreatic disease.** Endemic in **Florida, the Caribbean, and Hawaii**. Cats acquire it by eating infected lizards ("lizard poisoning"). **Treat with praziquantel.**
Hepatic lipidosis CAT	A cause of **Heinz body anemia**. The classic setting is an obese cat that stops eating for any reason — **nutritional support is the treatment**, delivered by feeding tube.

7.4 Diarrhea Localization

Feature	SMALL bowel	LARGE bowel
Volume	**Large volume** per defecation	**Small volume** per defecation
Frequency	**Normal to mildly increased**	**Markedly increased frequency**
Blood	**Melena** (digested, black)	**Hematochezia** (fresh red)
Mucus	Absent	**Present**
Tenesmus / urgency	Absent	**Present**
Weight loss / vomiting	**Common**	Uncommon

WHY THIS MATTERS ON THE EXAM

Histoplasmosis causes LARGE bowel diarrhea — increased frequency, decreased volume, hematochezia — and that fact is only usable if you know the localization table above. Likewise *Trichuris* (whipworm) and *Tritrichomonas* are large bowel; *Giardia* and IBD are typically small bowel.

7.5 Esophageal & Miscellaneous

Condition	Key Fact
Spirocerca lupi DOG	The **esophageal worm of dogs** causing **reactive granulomas** in the caudal esophageal wall. Can undergo **neoplastic transformation to osteosarcoma or fibrosarcoma**, and causes aortic aneurysms and spondylitis. Dung beetle intermediate host.
Metoclopramide	The antiemetic that **crosses the blood-brain barrier and antagonizes dopamine in the chemoreceptor trigger zone (CRTZ)**. Also a prokinetic. Contrast with maropitant (NK-1 antagonist) and ondansetron (5-HT_3 antagonist).
Dog and cat placentation	**ENDOTHELIOCHORIAL, ZONARY** placenta. Compare: **horses and pigs = epitheliochorial, diffuse** (maternal and fetal circulations remain completely separate — hence no transplacental antibody transfer and total dependence on colostrum). **Ruminants = synepitheliochorial, cotyledonary. Rabbits, rodents, primates = hemochorial, discoid** (most invasive).
Dentition	**Dog = 42** · **Cat = 30** · **Pig = 44** · **Horse = 36–44** · **Ruminant = 32**

8 | Urinary, Endocrine & Reproduction

8.1 Protein-Losing Nephropathy & Nephrotic Syndrome

Topic	Detail
PLN — the classic case	**A Soft-Coated Wheaten Terrier with distal limb edema and a history of weight loss.** (Also Shar Pei, Bernese Mountain Dog, Doberman.)
Where in the kidney	**The GLOMERULUS** — PLN is a glomerulonephritis (or amyloidosis). Tubular disease does not lose protein at this magnitude.
Treatment	**ACE inhibitors** such as benazepril or enalapril — they reduce glomerular capillary pressure and proteinuria. Add antithrombotics (loss of antithrombin III makes these patients hypercoagulable), a renal diet, and treat the underlying antigenic trigger.
Nephrotic syndrome — four components	1. **Hypoalbuminemia** · 2. **Proteinuria** · 3. **Hypercholesterolemia** · 4. **Edema / ascites** A complication of **glomerular disease**. (Protein-losing enteropathy produces hypoalbuminemia and edema too, but true nephrotic syndrome is by definition renal.)

8.2 Urolithiasis & Lower Urinary Tract

Topic	Detail
Most common feline stone at LOW urine pH CAT	**CALCIUM OXALATE.** Acidic urine favors oxalate; alkaline urine favors **struvite**. Calcium oxalate is **not** dissolvable — it must be removed surgically or by voiding urohydropropulsion.
Uroabdomen diagnosis	**Abdominal fluid : serum creatinine ≥ 2:1**; **abdominal fluid : serum potassium ≥ 1.4:1 (dog) / 1.9:1 (cat)**. See §4.4.
Hypospadias	An **abnormal placement of the urethral opening, ventral and caudal to the normal position** — a congenital defect of urethral fold fusion.

8.3 Diabetes Mellitus & Insulin Therapy

Concept	Detail
Insulins, shortest to longest acting	**Regular** (fastest — the DKA/CRI insulin) → **NPH / Vetsulin (lente)** — the dog insulin → **detemir / PZI** → **glargine** — the cat insulin → **ultralente**.
Interpreting a glucose curve — three things	**1. The nadir 2. The duration of effect 3. The clinical signs.** Never adjust insulin on a single glucose value; and always weigh the clinical signs above the curve.
Ideal nadir	**80–150 mg/dL in both dogs and cats.**
Somogyi phenomenon	**A period of hypoglycemia followed by rebound HYPERglycemia**, driven by counter-regulatory hormone release (glucagon, cortisol, epinephrine). **Seen when the insulin dose is TOO HIGH.** The trap: the owner reports a persistently high morning glucose and the reflex is to increase insulin — which makes it worse. **The correct action is to DECREASE the dose.** *The clue is an excessive fall in glucose during the curve, followed by a high reading.*
Stress hyperglycemia CAT	Cats develop marked stress hyperglycemia from restraint alone; below roughly **~290 mg/dL** stress remains a plausible explanation. **Use fructosamine** to distinguish transient stress hyperglycemia from true diabetes — it reflects the preceding 2–3 weeks.

8.4 Thyroid & Adrenal Notes

Topic	Detail
Euthyroid sick syndrome — three drugs	**Phenobarbital · Prednisone · Carprofen.** These suppress total T4 in a dog with normal thyroid function, producing a false diagnosis of hypothyroidism. Always confirm with free T4 by equilibrium dialysis plus TSH before treating.
Drug-induced hypoadrenocorticism	**Mitotane** (o,p'-DDD) causes adrenocortical necrosis and can produce iatrogenic **Addison's disease**. Trilostane can do the same reversibly.
Pseudo-Addison's DOG	***Trichuris vulpis* (whipworm)** causes GI sodium loss and potassium retention that mimics the hyponatremia/hyperkalemia of Addison's disease — with a **normal** ACTH stimulation test. See §13.

8.5 Reproduction — Estrus & Gestation Tables

Species	Estrus Duration	Cycle Type	Gestation
Dog	**9 days**	**Non-seasonally MONOESTRUS** — a single isolated heat that can occur at any time of year, roughly every 6–7 months	**62–65 days**
Cat	**7 days**	**Seasonally polyestrus** (anestrus in December); **induced ovulator**	**65 days**
Cow	**18 hours**	**Year-round polyestrus**; 21-day cycle	**~9 months (270 days)**
Mare	**5–7 days**	**Seasonally polyestrus** (anestrus in December); 21-day cycle	**11 months (>330 days)**
Sow	**40–60 hours**	**Year-round polyestrus**; 21-day cycle	**114 days** — "3 months, 3 weeks, 3 days"
Ewe / doe	**24 hours**	**Seasonally polyestrus**; cycle <20 days (~17)	**150 days**

THE ESTRUS AND CYCLE PATTERNS, CONDENSED

Estrus of 7 days: cats and horses.
21-day cycle: pigs, cattle, horses. (Sheep are shorter, ~17 days.)
Year-round polyestrus: pigs and cattle.
Seasonally polyestrus: sheep and goats (short-day), **horses and cats** (long-day — anestrus in December).
Non-seasonally monoestrus: the DOG.
Induced ovulators: cats, rabbits, ferrets, camelids.

8.6 Reproductive Miscellany

Topic	Detail
Estradiol in the dog	Causes two things: **bone marrow toxicity** (aplastic anemia — estrogen is myelotoxic in dogs and ferrets) and **pyometra** (estrogen primes the endometrium for progesterone-driven cystic hyperplasia).
Eclampsia	Periparturient hypocalcemia in the lactating bitch — tremors, hyperthermia, seizures. Treat with slow IV calcium gluconate **with ECG monitoring**, then wean or supplement the puppies.

Topic	Detail
Priapism	**Erection in the absence of sexual stimulation that does not resolve quickly.** Classically associated with **acepromazine** (especially in stallions, where it can be permanent).

9 | Dermatology

9.1 Dermatophytes

Organism	Host / Reservoir	Wood's Lamp	Morphology & Notes
Microsporum canis	**Cats are the natural host**; also dogs. The most common cause of dermatophytosis in dogs and cats.	**FLUORESCES apple-green** — the only one that reliably does, and only about 50% of the time.	**Macroconidia are spindle-shaped, thick-walled, with a paintbrush-like terminal knob, and contain MORE THAN 6 CELLS.**
Microsporum gypseum (and *M. persicolor*)	**Geophilic** — from soil. Digging dogs.	**Does NOT fluoresce**	Macroconidia are **<5 cells and thin-walled** — the discriminator from *M. canis*.
Trichophyton mentagrophytes Z	**Lives in RODENTS** — the zoonotic dermatophyte **spread from hamsters** and other small pets. Also rabbits and chinchillas.	**Does NOT fluoresce**	**Treat with terbinafine.**
Trichophyton verrucosum	**Cattle** — ringworm with thick gray-white crusts around the eyes and face in calves.	No	Highly zoonotic to handlers.

THE TWO MOST COMMON ZOOPHILIC DERMATOPHYTES

1. *Microsporum canis* — cats are the natural host; **fluoresces**.
2. *Trichophyton mentagrophytes* — rodents are the reservoir; **does not fluoresce**; treat with **terbinafine**.
A negative Wood's lamp never rules out ringworm. Culture on DTM is definitive.

Treatment by species

Patient	Drug
Dogs	**Ketoconazole** is acceptable in dogs.
Cats and small dogs	**ITRACONAZOLE. Avoid ketoconazole in cats** — poorly tolerated, causing anorexia, hepatotoxicity, and vomiting.
Rabbits, chinchillas, rodents	**Terbinafine.**

9.2 Mites

Mite	Morphology	Disease	Treatment
Sarcoptes Z	**BURROWING. Short legs with long unsegmented pedicles.** Round body.	**Intense, non-seasonal pruritus** — ear margins, elbows, hocks, ventrum. Positive pinnal-pedal reflex. **Zoonotic** (self-limiting in humans). Often only a few mites present, so **a negative skin scrape does not rule it out**.	Isoxazolines (afoxolaner, fluralaner), selamectin, ivermectin.
Demodex	**BURROWING** — lives in hair follicles. Cigar-shaped.	Usually **non-pruritic** alopecia unless secondarily infected. Juvenile-onset (hereditary) vs. adult-onset (look for underlying immunosuppression or endocrine disease).	Isoxazolines. Deep skin scrapes to diagnose.

Mite	Morphology	Disease	Treatment
Cheyletiella Z	**Surface / NON-burrowing. Prominent hook-like mouthparts.**	**"WALKING DANDRUFF"** — large scale over the **dorsal back** that appears to move. Affects small animals (and rabbits). **Pruritic** and **ZOONOTIC**.	**Isoxazolines**; selamectin, lime sulfur.
Otodectes cynotis	Surface, ear canal	Ear mites — dark, dry, coffee-ground exudate; intense head shaking. Most common in kittens.	**Imidacloprid/moxidectin** or selamectin — imidacloprid is the standard answer for mange in cats.
Chorioptes	**Surface. Long legs, short unsegmented pedicles.**	**Leg mange in horses and cattle**; distal limbs, perineum, tail head. **Seen in WINTER in the northeastern US.**	Topical sprays/dusts — because it lives on the surface.
Psoroptes	**Surface.** Long legs, **segmented** pedicles.	Common in cattle, sheep, and **rabbits** (*P. cuniculi* ear mite). **Reportable in sheep; eradicated from the US.**	Topical sprays/dusts.

BURROWING VS. SURFACE — THE CONCEPT BEHIND THE TREATMENT

BURROWING mites: *Sarcoptes* and *Demodex*. They live inside the skin, so they need **systemic** therapy and cause deep inflammation — hyperemia, papules, pustules.

SURFACE mites: *Psoroptes*, *Chorioptes*, *Cheyletiella*, *Otodectes*. They live on the skin surface, so **topical sprays and dusts work**.

Pedicle mnemonic: Chorioptes = **ch**ort pedicles, long legs. **Sarcoptes** = short legs, long pedicles.

10 | Oncology

10.1 Round Cell Tumors

THE ROUND CELL MNEMONIC — "T-LYMMPH"

TVT (transmissible venereal tumor) · **LYM**phoma · **M**ast cell tumor · **M**elanoma · **P**lasma cell tumor · **H**istiocytoma
Round cells are discrete, individualized cells with round nuclei that exfoliate readily on FNA — which is why cytology is so useful for this group and comparatively poor for mesenchymal tumors.

10.2 Tumors by Location and Species

Tumor	Key Facts
CNS tumors	**Meningioma is the most common BRAIN tumor in BOTH dogs and cats**, and the most common **SPINAL** tumor in **dogs**. **The most common SPINAL tumor in CATS is LYMPHOMA** — often FeLV-associated in young cats. That is the one exception in the grid, and it is what gets tested.
Mast cell tumor	**The most common cause of SPLENOMEGALY in CATS** — perform an FNA of the spleen. Visceral MCT in cats is a distinct, more aggressive entity than the cutaneous form. **Treatment in dogs: toceranib (Palladia)**, a receptor tyrosine kinase inhibitor targeting **c-KIT**. Surgery with wide margins remains first-line.
Feline mammary tumors CAT	**Roughly 85–90% are MALIGNANT** — feline mammary tumors are far worse than canine. **Size is the key prognostic indicator:** **> 3 cm → median survival time 4–6 months** after surgery **< 2 cm → median survival time ~3 years** Treatment is **bilateral radical mastectomy**. Early spay is strongly protective.
Canine mammary tumors DOG	**The classic "50/50 rule" — about half are benign and half malignant.** Risk is dramatically reduced by spaying before the first heat.
Osteosarcoma	**Negative prognostic indicator: HIGH ALP.** Elevated total or bone ALP at diagnosis correlates with shorter disease-free interval and survival. Large/giant breeds, metaphyseal ("away from the elbow, toward the knee"), highly metastatic to lung.
Basal cell tumor / adenoma	**Breeds predisposed: Himalayan and Persian cats.** Cytology: **large, round to polygonal cells with distinct borders**, often in tight clusters and pigmented.
Lymphoma and FeLV CAT	**Radiographic finding: a WIDENED MEDIASTINUM** — mediastinal lymphoma in a young FeLV-positive cat. Presents with dyspnea, non-compressible cranial thorax, and pleural effusion.
Limbal melanoma	**Slow growing with a low rate of metastasis.** See §6.3.

10.3 Feline Injection-Site Sarcoma

THE 3-2-1 RULE

Biopsy or excise a post-vaccination mass if it is:
3 — still present **3 months** after vaccination
2 — larger than **2 cm** in diameter
1 — still **increasing in size 1 month** after vaccination
Any **one** of the three triggers action. These tumors are locally aggressive and require radical excision — plan the surgery before you cut.

Topic	Detail
FVRCP	**F**eline **V**iral **R**hinotracheitis (herpesvirus-1) · **C**alicivirus · **P**anleukopenia.

Topic	Detail
Where to give it	**Distal to the right elbow.** The convention: **FVRCP right front, Rabies right rear, Leukemia left rear** — as distally as possible, so that a resulting sarcoma can be treated by amputation.

10.4 Chemotherapy Toxicities

Drug	Toxicity in DOGS	Toxicity in CATS
Doxorubicin	**CARDIOTOXIC** — cumulative, dose-dependent dilated cardiomyopathy. Also a severe vesicant on extravasation.	**NEPHROTOXIC.** Monitor renal values in cats, cardiac function in dogs.
Cisplatin	Nephrotoxic; requires saline diuresis.	**FATAL PULMONARY TOXICITY — never give cisplatin to a cat.** Causes fulminant pulmonary edema. Use carboplatin instead.
Azathioprine	Immunosuppressant; hepatotoxicity and myelosuppression possible.	**Contraindicated — cats lack TPMT (thiopurine methyltransferase)** and develop severe bone marrow suppression.
5-Fluorouracil	Used topically.	**Fatal neurotoxicity in cats.**

THE "NEVER IN CATS" DRUG LIST — WORTH MEMORIZING AS A BLOCK

Cisplatin (fatal pulmonary edema) · **5-FU** (fatal neurotoxicity) · **Azathioprine** (marrow suppression, no TPMT) · **Ketoconazole** (hepatotoxicity, poorly tolerated) · **Permethrin/pyrethroids** (tremors, seizures) · **Acetaminophen** (methemoglobinemia, no glucuronyl transferase) · **Phenol-based disinfectants**. The unifying theme for several of these: **cats are deficient in glucuronidation and several other conjugation pathways.**

RAPID-FIRE ONCOLOGY RECALL

Round cells → **T-LYMMPH**.
Splenomegaly in a cat → **mast cell tumor** → FNA the spleen.
Brain tumor, dog or cat → **meningioma**. Spinal tumor in a **cat** → **lymphoma**.
Widened mediastinum in a young cat → **FeLV mediastinal lymphoma**.
Feline mammary mass >3 cm → **MST 4–6 months**; <2 cm → **~3 years**.
High ALP with a bone tumor → **negative prognostic indicator for osteosarcoma**.
Post-vaccine mass → **3-2-1 rule**.
Doxorubicin → **cardiotoxic in dogs, nephrotoxic in cats**.

11 | Vector-Borne & Tick Disease

THE HIGHEST-YIELD SMALL ANIMAL INFECTIOUS BLOCK

Tick-borne disease is tested as a matched set: **organism → vector tick → target cell → geography → diagnostic finding → treatment.** Learn it as a grid, not as isolated facts.

11.1 The Master Table

Disease	Organism	Vector	Clinical Signs	Diagnosis & Treatment
Canine monocytic ehrlichiosis	***Ehrlichia canis***	***Rhipicephalus sanguineus*** — the brown dog tick. **NOT ornate, SHORT mouthparts.**	Four signs: **fever, anorexia, edema of the limbs, hypoalbuminemia.** Also thrombocytopenia, epistaxis, pancytopenia in the chronic phase (German Shepherds severely affected).	**Morulae in MONOCYTES.** Serology/PCR. **Doxycycline** 28 days.
Canine granulocytic ehrlichiosis	***Ehrlichia ewingii***	***Amblyomma*** — **ORNATE (white marking on the scutum) with LONG mouthparts.**	Polyarthritis, fever, lameness.	**Morulae in GRANULOCYTES (neutrophils).** Doxycycline.
Rocky Mountain spotted fever Z	***Rickettsia rickettsii***	***Dermacentor*** — the American dog tick. **ORNATE with SHORT mouthparts.**	**Pathogenesis is VASCULITIS** — fever, petechiae, edema, neurologic signs, thrombocytopenia. Acute and severe. **Endemic: south central and southern Atlantic states** (despite the name).	**Doxycycline. Key discriminator: RMSF is OVER IN 2 WEEKS** — it is an acute, self-limiting-or-fatal illness with no carrier state.
Cytauxzoonosis CAT	***Cytauxzoon felis***	***Amblyomma americanum*** — the **lone star tick**; ***Dermacentor*** is also implicated. The bobcat is the reservoir.	**The classic stem: fever, hepatosplenomegaly, lymphadenomegaly, and icterus in a cat in JULY in MISSOURI.** Four signs: **dark urine, icterus, fever, DIC and anemia.** **Infects MACROPHAGES and ERYTHROCYTES.** **Geography: Missouri and south central; Texas to Florida.**	**Cytology: "signet ring" / round piroplasms inside RBCs**, and **schizonts** in spleen, liver, blood, bone marrow, or lymph nodes. Giemsa or Diff-Quik. **Treat with ATOVAQUONE + AZITHROMYCIN.** **Often FATAL — treat as an emergency.**
Babesiosis DOG	***Babesia canis*** (large) and ***B. gibsoni*** (small)	***Rhipicephalus* (Boophilus)**. *B. gibsoni* also transmitted by **dog bites and transfusion**.	**RBC destruction → hemolytic ANEMIA and THROMBOCYTOPENIA** in dogs. **Breeds predisposed: Greyhounds and Pit Bulls** — Pit Bulls specifically for *B. gibsoni* via fighting/bite transmission. Seen in **dogs and horses**.	***B. canis*** **→ IMIDOCARB DIPROPIONATE.** ***B. gibsoni*** **→ ATOVAQUONE + AZITHROMYCIN** (imidocarb is poorly effective against the small *Babesia*).

Disease	Organism	Vector	Clinical Signs	Diagnosis & Treatment
Lyme disease Z	***Borrelia burgdorferi***	***Ixodes*** — the deer tick. **Reservoir: the white-footed mouse.**	**In DOGS, four signs: arthritis (shifting-leg lameness), lymphadenopathy, fever, anorexia.** Also Lyme nephritis (Labradors, Golden Retrievers) — a protein-losing nephropathy that is often fatal. **In HUMANS: a cutaneous rash — erythema migrans** (the "bull's-eye").	**Antibody against the C6 peptide indicates NATURAL EXPOSURE** (as opposed to vaccination) — that is the point of the C6 test. **Doxycycline.**
Anaplasmosis	***Anaplasma phagocytophilum***	***Ixodes*** — same tick as Lyme, so co-infection is common.	**Fever, lameness, thrombocytopenia, and NO anemia.**	**Morulae in NEUTROPHILS.** Doxycycline.
Hemotropic mycoplasmosis CAT	***Mycoplasma haemofelis***	Fleas; also bites and transfusion.	**A cat with fever and cocci in chains/rings ON the red cell surface.** Regenerative hemolytic anemia. **Most common in outdoor male cats.**	**Organisms ON the surface of RBCs** — not inside them. **Doxycycline** ± prednisolone.
Bartonellosis Z	***Bartonella henselae*** (cat) and ***B. vinsonii*** (dog)	**FLEAS are the main reservoir for feline infection** — transmitted between cats by flea feces, and to humans by scratch.	***B. henselae* = "CAT SCRATCH FEVER"** in humans — regional lymphadenopathy. Cats are usually asymptomatic carriers. ***B. vinsonii* in dogs → INFECTIOUS ENDOCARDITIS.** The stem: **a dog on no flea/tick preventive with a new heart murmur.**	Doxycycline or azithromycin; flea control is the real answer.
Tularemia Z	***Francisella tularensis***	Ticks; also direct contact and aerosol.	**Dogs and cats are exposed by hunting RODENTS or RABBITS that have been bitten by ticks.** Cats are more susceptible than dogs — fever, lymphadenopathy, oral ulcers.	**Highly ZOONOTIC with a very low infectious dose — a recognized BIOLOGICAL WARFARE / select agent.** Handle suspect cases with extreme caution and warn the lab. Doxycycline or aminoglycosides.
Hepatozoonosis DOG	***Hepatozoon americanum***	**Gulf Coast tick** — acquired by **INGESTING** the tick, not by its bite.	Severe muscle pain, stiff gait, fever, marked neutrophilia.	**Radiographs show striking PERIOSTEAL BONE PROLIFERATION** along the long bones — the pathognomonic finding.
Leishmaniasis Z	*Leishmania infantum*	**SAND FLY** (*Lutzomyia*/ *Phlebotomus*). Also vertical transmission in Foxhounds in the US.	Visceral and cutaneous disease — weight loss, exfoliative dermatitis, onychogryphosis, PLN.	Zoonotic. Poor cure rate; allopurinol + antimonials.
American trypanosomiasis (Chagas) Z	***Trypanosoma cruzi***	**Kissing bug** (*Triatoma*) — infection occurs via the bug's **feces**, not its bite.	**Myocarditis and megaesophagus / megacolon.** Young dogs in Texas and the Gulf states; sudden death or right-sided heart failure.	Zoonotic. No reliably effective treatment.

Disease	Organism	Vector	Clinical Signs	Diagnosis & Treatment
Heterobilharzia americana DOG	A schistosome (blood fluke)	Snail intermediate host; water contact in the Gulf states.	**Migrates across the intestinal wall causing GRANULOMATOUS inflammation**, hepatic granulomas, and **hypercalcemia**.	Fecal saline sedimentation (not flotation) or PCR. Praziquantel + fenbendazole.

11.2 Tick Identification — The Two Questions That Answer Everything

ORNATE OR NOT? LONG OR SHORT MOUTHPARTS?

Amblyomma — **ORNATE** (white marking on the scutum) + **LONG** mouthparts → ***E. ewingii*, cytauxzoonosis, heartwater**. *A. americanum* = lone star tick.
Dermacentor — **ORNATE** + **SHORT** mouthparts → **RMSF**, tick paralysis. The American dog tick.
Rhipicephalus — **NOT ornate** + **SHORT** mouthparts → ***E. canis*, *Babesia***. The brown dog tick, and the one that infests houses and kennels.
Ixodes — small, dark, no ornamentation → **Lyme and *Anaplasma***. Deer tick.
Soft ticks (Argasidae): *Otobius megnini* (spinose ear tick — predilection for the **ears** of cattle), ***Ornithodoros coriaceus*** (epizootic bovine abortion), ***Argas persicus*** (fowl spirochetosis).

11.3 The Blood-Parasite Discriminators

FOUR ONE-LINE STEMS — MEMORIZE THESE VERBATIM

Anemia + organisms ON the red cells → *Mycoplasma haemofelis*
Fever/lameness + thrombocytopenia + NO anemia + morulae in NEUTROPHILS → *Anaplasma phagocytophilum*
Cat crashing fast + signet-ring inclusions INSIDE RBCs + southeastern US + tick exposure → *Cytauxzoon felis* — the emergency, and the one that kills cats quickly if missed
Morulae in MONOCYTES + limb edema + hypoalbuminemia → *Ehrlichia canis*

THE TREATMENT SHORTCUT

Doxycycline covers essentially every **rickettsial and bacterial** tick-borne disease: *Ehrlichia*, *Anaplasma*, RMSF, Lyme, *Mycoplasma haemofelis*, *Bartonella*.
The **protozoal** ones need something else: ***Babesia canis* → imidocarb**; ***B. gibsoni* and *Cytauxzoon* → atovaquone + azithromycin**.

12 | Systemic Mycoses

LEARN THESE FOUR BY GEOGRAPHY + CYTOLOGY

Every systemic mycosis question can be answered from two facts: **where the dog lives** and **what the organism looks like on cytology**. The treatment is almost always **itraconazole**.

Organism	Geography	Clinical Signs	Cytology & Treatment
Blastomycosis *Blastomyces dermatitidis*	**Ohio, Mississippi, and Missouri river valleys** (and the Great Lakes / St. Lawrence). Sandy, acidic soil near water.	**"BELLS"** in dogs: **B**one · **E**yes (uveitis) · **L**ymph node enlargement · **L**ungs (cough, diffuse nodular pattern) · **S**kin lesions	**BROAD-BASED BUDDING YEAST**, thick double-contoured wall, ~8–20 µm. **Itraconazole for several months (at least 60 days)** — usually 3–6 months and at least 1 month past resolution.
Histoplasmosis *Histoplasma capsulatum*	**Same river valleys as blasto. Soil enriched with BAT or BIRD droppings — CATS are at high risk.**	**LARGE BOWEL DIARRHEA** — increased frequency, decreased volume, hematochezia (dogs). Disseminated wasting, hepatosplenomegaly, and respiratory disease in cats.	**SMALL INTRACELLULAR yeast** (2–4 µm) **with a basophilic center, inside macrophages.** **Diagnosis: submit URINE for fungal antigen.** Itraconazole.
Coccidioidomycosis *Coccidioides immitis* ("Valley fever")	**WEST COAST** — the desert southwest: Arizona, California's San Joaquin Valley, New Mexico, west Texas.	**DOGS:** cough, **lameness** (osteomyelitis), draining tracts, pneumonia, lymphadenopathy. **CATS: skin lesions** predominantly.	**SPHERULES** — large, double-walled structures containing **endospores**. Highly distinctive. **Itraconazole** (avoid ketoconazole in cats). **Treat for 6–12 months** — the longest course of any of these.
Cryptococcosis *Cryptococcus neoformans* CAT	Worldwide. **Associated with PIGEON DROPPINGS.**	**"ROMAN NOSE" in a cat** — a firm swelling over the bridge of the nose from granulomatous rhinitis, often with a polyp protruding from the nostril. Also CNS disease and chorioretinitis.	**Small yeast with a LARGE clear capsule and NARROW-BASED budding.** India ink or mucicarmine. **Diagnosis: LATEX AGGLUTINATION** for capsular antigen — a superb test. Fluconazole (best CNS penetration) or itraconazole.
Aspergillosis *Aspergillus fumigatus* DOG	Ubiquitous environmental mold.	**Nasal aspergillosis in DOLICHOCEPHALIC dogs — the German Shepherd is the classic breed.** Chronic ulcerated nasal discharge, depigmentation of the nasal planum, turbinate destruction and pain.	**Branching, septate fungal HYPHAE** on histopathology. **Treat with CLOTRIMAZOLE** — infused topically into the nasal cavity/frontal sinus under anesthesia. **Post-recovery complication: SEVERE LARYNGEAL EDEMA** — a real anesthetic-recovery risk after nasal infusion. Have intubation supplies ready.
Sporotrichosis *Sporothrix schenckii* Z	Worldwide; soil and plant material.	**A huge, ulcerated, non-healing lesion on a CAT's NOSE**, or nodules along lymphatics after a puncture wound.	**Extremely ZOONOTIC from cats** — cats carry an enormous organism burden and transmit through intact skin. **Wear gloves.** Itraconazole.

BLASTO VS. HISTO — THE DIRECT COMPARISON

HISTO IS SMALLER, and it is **INTRACELLULAR inside macrophages** with a **basophilic center**.
BLASTO IS LARGER, **extracellular**, with **BROAD-BASED budding** and a thick refractile wall.
Both live in the **same river valleys** — so geography will not separate them. **Size and location relative to the cell will.**
Add the third: **Crypto** = **narrow-based** budding with a **huge capsule**.

RAPID-FIRE MYCOLOGY RECALL

Broad-based budding yeast, dog with bone/eye/lung/skin/node disease → **blastomycosis** → BELLS.
Small intracellular yeast in macrophages + large bowel diarrhea → **histoplasmosis** → **urine antigen**.
Spherules with endospores, dog in Arizona, lame → **coccidioidomycosis** → treat 6–12 months.
Cat with a swollen nose bridge → **cryptococcosis** → **latex agglutination**, pigeon droppings.
German Shepherd with destructive nasal discharge → **aspergillosis** → **clotrimazole infusion**, watch for laryngeal edema on recovery.
Nasty ulcerated lesion on a cat's nose → **sporotrichosis** → **gloves, highly zoonotic**.
When in doubt on treatment → **itraconazole**.

13 | Parasitology

THE FOUR CLASSES OF WORMS

Hookworms (nematode) · **Roundworms** (nematode) · **Whipworms** (nematode) · **Tapeworms** (cestode).
Three of the four are nematodes — which is why **fenbendazole** handles most of them and **praziquantel** is reserved for the cestodes and trematodes.

13.1 Nematodes

Parasite	Host & Location	Disease	Diagnosis & Treatment
Ancylostoma caninum (hookworm) Z	Small intestine of dogs.	**In DOGS: ANEMIA** — a blood-feeding worm; severe iron-deficiency anemia in puppies. **In HUMANS: CUTANEOUS LARVAL MIGRANS** — serpiginous, intensely itchy tracks in the skin where larvae penetrate but cannot mature.	**Egg: thin-walled with 2–8 cells.** Pyrantel or fenbendazole.
Toxocara canis (roundworm) Z	Dogs. (***Toxocara cati*** is the feline equivalent.)	**In DOGS: COUGH IN A YOUNG PUPPY** — from the tracheal migration phase — plus pot belly, poor doing, vomiting worms. **In HUMANS: VISCERAL and OCULAR LARVAL MIGRANS** — a leading infectious cause of childhood blindness.	**Transmission: TRANSPLACENTAL in puppies; TRANSMAMMARY in kittens.** **Pyrantel** or fenbendazole. Deworm puppies starting at 2 weeks.
Toxascaris leonina	Dogs and cats.	**The roundworm that is NOT zoonotic** — the one exception in the group.	Memory hook: ***Toxocara canis* is the one that "leaves" the gut and migrates (zoonotic, causes ocular larval migrans); *Toxascaris leonina* stays put.**
Trichuris vulpis (whipworm) DOG	**Cecum and colon** of dogs. (***Trichuris suis*** is the porcine whipworm, also in the **cecum**.)	**Large bowel diarrhea with hematochezia.** Two classic complications: **1. CECOCOLIC INTUSSUSCEPTION** **2. PSEUDO-ADDISON'S DISEASE** — hyponatremia and hyperkalemia mimicking hypoadrenocorticism, but with a **normal ACTH stimulation test**.	**FENBENDAZOLE.** Eggs are shed intermittently — a negative fecal does not rule it out. Long environmental survival.
Strongyloides stercoralis ("threadworm") Z	Small intestine.	**ANEMIA and MUCOID DIARRHEA in kittens and puppies.** Skin penetration causes dermatitis.	**BAERMANN fecal technique** — because you are looking for **larvae**, not eggs. Ivermectin/fenbendazole.
Spirocerca lupi DOG	**Esophagus** of dogs.	**Reactive esophageal granulomas** that may transform into **sarcoma**. Also aortic aneurysm, spondylitis.	Dung beetle intermediate host. Prolonged doramectin/milbemycin.
Ollulanus tricuspis CAT	**Stomach of CATS only.**	**Chronic VOMITING** and gastritis.	Examine **vomitus**, not feces.
Gnathostoma ("stomach worm") Z	Stomach wall.	**Usually vomited up; causes gastritis.**	**Humans acquire it by ingesting UNDERCOOKED FISH** → migratory cutaneous swellings and, rarely, CNS disease. **Albendazole.**

Parasite	Host & Location	Disease	Diagnosis & Treatment
Capillaria aerophila	**Lungs / airways.**	Chronic cough.	**Egg: ASYMMETRIC with TERMINAL PLUGS (bipolar plugs)** — a whipworm-like egg but found on a lung sample. **Fenbendazole for 10 days** (same protocol as *Trichuris* in a dog).
Aelurostrongylus abstrusus CAT	**Lungworm diagnosed ONLY in cats.**	Cough, wheezing — a mimic of feline asthma.	**Baermann** for L1 larvae. Fenbendazole.

THE ZOONOTIC LARVAL MIGRANS QUESTIONS — GET THE DIRECTION RIGHT

CUTANEOUS larval migrans → *Ancylostoma* (HOOKWORM) — skin.
VISCERAL and OCULAR larval migrans → *Toxocara canis* (ROUNDWORM) and ***Baylisascaris*** (raccoon roundworm — causes devastating neural larva migrans).
Mnemonic: **hook**worm **hook**s into the skin; **round**worm goes **round** the body to the organs and eye.

13.2 Cestodes (Tapeworms)

Tapeworm	Key Features	Treatment
Dipylidium caninum	**Proglottids in the feces look like WHITE RICE** (or cucumber seeds) and move. **Intermediate host: the FLEA** (and chewing lice) — the animal must swallow a flea to become infected.	**Praziquantel** — plus **flea control**, or it will simply recur.
Taenia taeniaeformis / *T. pisiformis*	Acquired by eating rodents or rabbits (the intermediate host).	**Praziquantel.**
Echinococcus granulosus Z	**The DOG is the definitive host** and passes eggs in feces. **In HUMANS it causes HYDATID CYST DISEASE — potentially fatal.** Cysts form in the **liver and lungs**; in the intermediate hosts (sheep) the same. The tiny adult tapeworm causes no disease in the dog — **the danger is entirely to the people around it.**	**Praziquantel.** A genuine public health priority in sheep-raising regions; do not feed raw offal to dogs.

13.3 Trematodes (Flukes)

Fluke	Location & Disease	Diagnosis & Treatment
Paragonimus kellicotti	**The LUNG fluke of dogs and cats.** Cystic lung lesions, cough, pneumothorax.	**Two intermediate hosts: the SNAIL and the CRAYFISH.** **Egg has a SINGLE OPERCULUM** (a distinct "lid"). **Fenbendazole or praziquantel.**
Platynosomum fastosum CAT	**The PANCREATIC and HEPATOBILIARY fluke of cats.** Florida, Caribbean, Hawaii. From eating lizards.	**Praziquantel.**
Heterobilharzia americana	Blood fluke; **granulomatous** intestinal and hepatic disease, hypercalcemia.	Saline sedimentation or PCR; praziquantel + fenbendazole.
General rule	**Flukes are treated with PRAZIQUANTEL.**	

13.4 Protozoa

Protozoan	Identification	Disease & Treatment
Giardia Z	**TWO nuclei** outlined by adhesive discs; **pear-shaped, binucleate**; swims in a **"falling leaf" motion**.	**Diagnose by trophozoites on a direct fecal smear** (plus fecal ELISA/zinc sulfate flotation). Small bowel diarrhea. **Treat with fenbendazole or metronidazole.**
Tritrichomonas foetus CAT	**Looks like *Giardia* BUT has only ONE nucleus and an UNDULATING MEMBRANE.**	**Seen in KITTENS with chronic, unresponsive large bowel diarrhea** — often young cats from catteries or shelters. **Treat with RONIDAZOLE** (not metronidazole — that is the trap).
Cystoisospora (*Isospora*)	**2 sporocysts.**	**Infects dogs, cats, and pigs** — a major cause of severe diarrhea in suckling pigs. **The two most common coccidia of CATS: *Cystoisospora felis* and *C. rivolta*** — think foul-smelling diarrhea. **Treat with sulfadimethoxine (Albon).**
Eimeria	**4 sporocysts.**	**Infects ruminants, horses, rabbits, and poultry — NOT dogs and cats.** A dog or cat that has eaten **rabbit feces** will pass *Eimeria* oocysts spuriously — a **pseudoparasite**, not an infection.
Cryptosporidium Z	**Requires ACID-FAST staining or IFA** to find — the oocysts are tiny (4–6 µm) and easily missed.	**RESISTANT TO STANDARD DISINFECTANTS** and to chlorination — which is why it causes **waterborne human outbreaks**. Zoonotic. **Treat with azithromycin, tylosin, or clindamycin** (no reliably effective drug; supportive care in immunocompetent hosts).
Toxoplasma gondii Z	Tissue cysts; tachyzoites.	**The CAT is the definitive host** — only cats shed oocysts, and only for a brief period after a first infection. **Oocysts require 1–5 DAYS after shedding to SPORULATE and become infective** — which is exactly why **daily litter box cleaning prevents transmission**. **In infants infected in utero: CHORIORETINITIS and mental retardation.** **Toxoplasmosis does NOT cause abortion in COWS** (it is a major abortifacient in sheep and goats). **Treat with CLINDAMYCIN** (also used for *Neospora caninum*; TMS is an alternative).

COCCIDIA VS. *GIARDIA* — AND *ISOSPORA* VS. *EIMERIA*

Coccidia are **intracellular** protozoa that destroy the intestinal epithelium — *Eimeria*, *Cystoisospora*.
Giardia is an **extracellular** flagellate that coats and blunts the villi without invading.
***Eimeria* = 4 sporocysts = livestock, rabbits, poultry.**
***Isospora/Cystoisospora* = 2 sporocysts = dogs, cats, pigs.**

TROPHOZOITES BEST FOUND ON A DIRECT SMEAR

Direct fecal smear (fresh, warm, saline) is the technique for motile organisms that **flotation destroys**: ***Giardia***, ***Tritrichomonas***, ***Entamoeba***, and ***Balantidium***. If a question asks how to find a wiggling protozoan, the answer is a direct smear — not a float.

13.5 Antiparasitic Drug Map

Drug	Spectrum
Pyrantel	**Hookworms and roundworms.** Safe in very young puppies and kittens.
Fenbendazole	***Trichuris, Giardia, Capillaria, Paragonimus, Strongyloides***, roundworms and hookworms. **10 days for *Capillaria aerophila* and *Trichuris* in a dog.**
Praziquantel	**ALL tapeworms (cestodes) and flukes (trematodes)** — *Dipylidium*, *Taenia*, *Echinococcus*, *Paragonimus*, *Platynosomum*. In horses and small animals alike.
Ivermectin / macrocyclic lactones	Nematodes and arthropods. **NOT effective against tapeworms/cestodes.** Good for strongyles and heartworm prevention.
Metronidazole	*Giardia*, *Entamoeba*, anaerobes.
Ronidazole	***Tritrichomonas foetus*** — the specific drug.
Sulfadimethoxine (Albon)	***Cystoisospora*** coccidiosis.
Clindamycin	***Toxoplasma*** and *Neospora*; also *Cryptosporidium*, anaerobes.
Albendazole	***Gnathostoma***; nematodes and flukes in cattle and camelids.
Isoxazolines (afoxolaner, fluralaner)	***Cheyletiella, Sarcoptes, Demodex***, fleas, ticks.
Imidacloprid	**Mange and ear mites in cats** (typically as imidacloprid/moxidectin).

TWO PROTOZOAL DRUG CLASSES WORTH NAMING

The **two antibiotic classes with heavy activity against both protozoa and bacteria** are **the nitroimidazoles** (metronidazole, ronidazole, tinidazole) and **the sulfonamides** (including potentiated sulfas). Add **the lincosamides** (clindamycin) as a third with real antiprotozoal activity.

RAPID-FIRE PARASITOLOGY RECALL

White rice in the feces → ***Dipylidium*** → **praziquantel + flea control**.
Coughing puppy → ***Toxocara canis*** migration.
Cutaneous larval migrans → **hookworm**. Visceral/ocular → ***Toxocara***.
Cecocolic intussusception or pseudo-Addison's → ***Trichuris vulpis*** → **fenbendazole**.
Kitten with unresponsive diarrhea → ***Tritrichomonas foetus*** → **ronidazole**.
One nucleus + undulating membrane → ***Tritrichomonas***. Two nuclei + falling leaf → ***Giardia***.
Baermann → **larvae** → ***Strongyloides, Aelurostrongylus***.
Single-operculate egg + crayfish → ***Paragonimus***.
Acid-fast tiny oocyst, chlorine-resistant → ***Cryptosporidium***.
Dog definitive host, hydatid cysts in people → ***Echinococcus granulosus***.
Oocysts sporulate in **1–5 days** → clean the litter box **daily**.

14 | Feline-Specific Medicine

WHY CATS GET THEIR OWN CHAPTER

Cats are not small dogs, and the NAVLE knows it. Nearly every feline question turns on one of four physiologic facts: **deficient glucuronidation** (drug toxicities), **preformed blood-type antibodies** (transfusion), **fused pancreatic and bile ducts** (triaditis), and **obligate carnivore nutrient requirements** (taurine, arginine, thiamine).

14.1 Respiratory

Condition	Key Facts
Feline asthma	**Three radiographic hallmarks:** **1. HYPERINFLATION** — mucus plugging and inflammation cause air trapping; look for a flattened diaphragm. **2. BRONCHIAL PATTERN** — irregularly thickened bronchial walls seen end-on as **"donuts"** and in profile as **"train tracks."** **3. RIGHT MIDDLE LUNG LOBE CONSOLIDATION** — from mucus plugging of that lobe's bronchus. Radiographically this reads as a **diffuse bronchial to bronchointerstitial pattern with pulmonary hyperinflation**. **Treatment: terbutaline** — a **beta-2 adrenergic agonist** that bronchodilates and relaxes the airways — plus inhaled or systemic corticosteroids.
Pleural effusion — two most common causes	**1. Heart failure 2. Neoplasia.** (Then pyothorax, chylothorax, and FIP.)
Rivalta test	A cage-side test to separate transudate from exudate, classically used in **FIP** workups. **Method:** add one drop of 98% acetic acid to 8 mL of distilled water in a clear tube, then carefully layer **one drop of the effusion** on top. **NEGATIVE:** the drop **dissolves** and disperses. **POSITIVE:** the drop **keeps its shape** — remains a ball or drifts down as a string — indicating a high protein and inflammatory mediator content.

14.2 Feline Infectious Disease

Agent	Key Facts
FeLV	**ELISA and PCR test for CIRCULATING ANTIGEN — not antibody.** That is why a positive test in a kitten may still clear, and why antibody-based logic does not apply. **Radiographs: widened mediastinum from mediastinal lymphoma.**
Feline herpesvirus-1 (FVR)	**Conjunctivitis and keratitis**, dendritic corneal ulcers, chronic upper respiratory disease with lifelong latency and stress-induced recrudescence. **Treat with FAMCICLOVIR** (plus L-lysine, though evidence is weak).
Chlamydia felis	**Suggested by INTRACYTOPLASMIC INCLUSION BODIES in conjunctival/eye discharge cytology.** Predominantly a **conjunctivitis** (chemosis) rather than a true URI. **Treat with ORAL DOXYCYCLINE** — systemic therapy is required because topicals do not clear the carrier state. **Terramycin (oxytetracycline)** is an alternative for *Chlamydia* or *Mycoplasma*.
Cytauxzoonosis	See §11.1. **Fever, hepatosplenomegaly, lymphadenomegaly, icterus, in Missouri, in July.** Atovaquone + azithromycin. Usually fatal.
Mycoplasma haemofelis	**Cocci in rings ON the RBC surface**, fever, hemolytic anemia. **Most common in outdoor male cats.** Doxycycline.
Toxoplasmosis	Cat is the definitive host; oocysts sporulate in **1–5 days**. Clindamycin.

14.3 Feline Odds and Ends

Topic	Detail
Feline acne	An **IDIOPATHIC** condition — comedones on the chin. Manage with hygiene, plastic bowl avoidance, and topical therapy.
Most common brain tumor	**Meningioma** (surgically resectable, good prognosis in cats).
Most common spinal tumor	**LYMPHOMA** — the exception to the meningioma rule.
Most common cause of splenomegaly	**Mast cell tumor** — FNA the spleen.
Most common congenital heart disease	**VSD.**
Most common urolith at low pH	**Calcium oxalate.**
Stress hyperglycemia threshold	Around **~290 mg/dL**; confirm true diabetes with **fructosamine**.
Anemia of chronic disease	**Mild, non-regenerative, normocytic, normochromic** — driven by inflammatory cytokines. The most common anemia in a sick cat.
Mange treatment	**Imidacloprid.**
Cholangitis antibiotic	**Amoxicillin-clavulanic acid.**
Ringworm treatment	**Itraconazole** — **never ketoconazole** in cats.
Petroleum distillate toxicity	**Buprenorphine** for analgesia; do not induce emesis.

15 | Clinical Pharmacology, Anesthesia & Analgesia

15.1 Antimicrobial Side Effects — The Tested Set

Drug / Class	Adverse Effect
Aminoglycosides (gentamicin, amikacin)	Three toxicities to name together: **OTOTOXIC · NEPHROTOXIC · NEUROMUSCULAR BLOCKADE.** Nephrotoxicity is worst in dehydrated patients — **never give an aminoglycoside to a volume-depleted animal**. Once-daily dosing reduces renal accumulation.
Tetracyclines	**Hepatotoxicity and potential nephrotoxicity — CONTRAINDICATED in renal insufficiency.** Also cause **discoloration and skeletal/dental abnormalities in the young**, and **primary photosensitization**. **DOXYCYCLINE is the tetracycline of choice in renal compromise** (eliminated largely by the GI tract, not the kidney) and **causes fewer skeletal abnormalities**.
Doxycycline in HORSES	**Small IV doses in horses are associated with cardiac arrhythmias, collapse, and DEATH.** Oral doxycycline is acceptable in horses; **IV doxycycline is not.**
Sulfonamides / TMS	The second "toxic T." Causes **KCS (keratoconjunctivitis sicca)**, **type I and type III hypersensitivity**, **hepatitis**, **hemolytic anemia**, and **bone marrow suppression at high doses**. Also polyarthritis and fever. **Sulfa drugs cause a type III hypersensitivity specifically in DOBERMAN PINSCHERS** — a well-recognized breed idiosyncrasy.
Lincosamides (clindamycin)	**GI upset.** Esophageal stricture in cats if dosed dry — always follow with water or food.
Macrolides (erythromycin)	**GI problems — in HORSES, FATAL COLITIS.** Note the related trap: when a **foal** is treated with erythromycin for *Rhodococcus*, **the MARE is at risk of *C. difficile* enterocolitis** from grooming the foal.
Macrolides and cephalosporins	**Cause pain and swelling on IM injection.**
Chloramphenicol	Reversible bone marrow suppression in animals; **idiosyncratic irreversible aplastic anemia in HUMANS** — counsel owners to wear gloves.

THE "TWO TOXIC TS"

Tetracycline — hepatotoxic, nephrotoxic, avoid in renal insufficiency, photosensitization, skeletal changes in the young.
TMS (trimethoprim-sulfa) — KCS, type I and III hypersensitivity, hepatitis, hemolytic anemia, marrow suppression.

15.2 Antimicrobial Spectrum

Question	Answer
Which two have NO anaerobic activity?	**Fluoroquinolones (enrofloxacin/Baytril) and aminoglycosides.** *Both require oxygen-dependent uptake or aerobic metabolism — which is precisely why they fail in abscesses, necrotic tissue, and any anaerobic environment.*
Which three are BEST for anaerobes?	**Penicillins · Clindamycin · Metronidazole.** (Metronidazole is the most reliably anaerobicidal.)
Which two classes hit protozoa AND bacteria?	**Nitroimidazoles** (metronidazole, ronidazole) and **sulfonamides** (including potentiated sulfas). Clindamycin also qualifies.
Which drugs are ototoxic? (four)	**Chlorhexidine · Alcohols · Aminoglycosides · Polymyxins.** *This is why you must confirm an intact tympanic membrane before putting anything in an ear canal.*
Which two drugs are nephrotoxic?	**Aminoglycosides — gentamicin and amikacin.**

15.3 Otic Therapeutics

Agent	Role
Dexamethasone / corticosteroids	The common anti-inflammatory placed in the ear to **reduce inflammation and canal swelling** — often the single most important component.
Enrofloxacin (otic Baytril)	Treats **gram-negative rods, especially *Pseudomonas*** — the difficult chronic otitis pathogen.
Combination products (e.g. Otomax, Mometamax)	A single product containing an **antibacterial, a steroid, and an antifungal** — because most otitis externa is a mixed *Malassezia*/bacterial infection with an allergic underlying cause.

15.4 Cardiovascular & Emergency Drugs

Drug	Mechanism	Use
Phenylephrine	**Alpha-1 agonist**	Vasoconstriction and increased blood pressure. **Norepinephrine and dopamine also produce vasoconstriction via alpha-1.**
Dobutamine and dopamine	**BETA-1 receptors**	Increase **cardiac contractility** (positive inotropes).
Epinephrine in CPR	Alpha-1 mediated peripheral vasoconstriction	**Increases CORONARY (and cerebral) perfusion** during chest compressions — that is the reason it is given, not the beta effect.
Atropine	**Muscarinic antagonist** — parasympatholytic	**Increases heart rate.** Also for organophosphate toxicosis and as a mydriatic.
Lidocaine and mexiletine	**Class IB antiarrhythmics**	Ventricular arrhythmias. **Mexiletine is the oral drug for longer-term management** (can cause GI upset). Lidocaine is the acute IV drug.
Diltiazem	Calcium channel blocker	**Atrial fibrillation, supraventricular tachycardia, HCM.**
Amlodipine	Calcium channel blocker	**Hypertension** — first-line in cats.
Enalapril / benazepril	ACE inhibitors	CHF, proteinuria/PLN, hypertension.
Spironolactone	**Aldosterone antagonist**, potassium-sparing	Adjunct in CHF; competitive antagonist in the ADH/aldosterone axis.
Furosemide	Loop diuretic	**First drug in acute CHF**, including the cat (IV or IM).

15.5 Anesthesia & Sedation

Drug	Mechanism & Notes
Acepromazine	**Phenothiazine tranquilizer — antagonizes central DOPAMINE receptors.** No analgesia, not reversible, causes **hypotension** via alpha-1 blockade, and is associated with **PRIAPISM** (notably in stallions). Historically avoided in seizure patients, though that concern is now largely disproven.
Ketamine	**Antagonizes NMDA (N-methyl-D-aspartate) receptors**, decreasing glutamate-mediated excitation and central sensitization. Dissociative anesthesia; preserves airway reflexes; excellent for wind-up pain.
Benzodiazepines (diazepam, midazolam)	**Enhance GABA-A receptor activity** (increase chloride conductance → hyperpolarization). Anxiolysis, muscle relaxation, anticonvulsant. **Diazepam per rectum** for status epilepticus.
Xylazine and **detomidine**	**Alpha-2 adrenergic agonists.** Sedation, analgesia, muscle relaxation — and **HYPERGLYCEMIA** (alpha-2 stimulation inhibits insulin release). Reversible with atipamezole/yohimbine.
Succinylcholine	**Depolarizing neuromuscular blocker** — used with isoflurane during **lung lobectomy** to keep the patient absolutely still. **No analgesia and no sedation** — the patient must be fully anesthetized.
Dantrolene	**RYANODINE RECEPTOR ANTAGONIST** — blocks calcium release from the sarcoplasmic reticulum. **In PIGS: prevents/treats MALIGNANT HYPERTHERMIA** (porcine stress syndrome). **In HORSES: post-anesthetic myositis** and exertional rhabdomyolysis.

Drug	Mechanism & Notes
Doxapram	Central respiratory stimulant — **neonatal resuscitation**.
Methocarbamol	Centrally acting muscle relaxant — the antidote-of-convenience for **tremorgenic toxicoses**: pyrethroids, strychnine, metaldehyde.

15.6 Analgesia

Concept	Detail
Drugs with a CEILING EFFECT	**Buprenorphine** (a partial mu agonist) and **butorphanol** (a kappa agonist / mu antagonist), plus the **NSAIDs** as a class. **A ceiling effect means increasing the dose does not increase analgesia — it only increases side effects.** For severe pain you must switch to a full mu agonist (morphine, hydromorphone, methadone, fentanyl), not escalate the dose.
COX-2 selective anti-inflammatories	**Firocoxib** and **deracoxib** (the "-coxib" suffix). Spare the constitutive COX-1 prostaglandins that maintain gastric mucosa and renal perfusion.
Buprenorphine	Also the analgesic of choice for **petroleum distillate toxicity in cats**. Excellent transmucosal absorption in cats.

15.7 Immunosuppressants & Miscellaneous

Drug	Notes
Azathioprine	**Immunosuppressant** used for IMHA and ITP in dogs. **TOXIC TO CATS — they lack THIOPURINE METHYLTRANSFERASE (TPMT)** and cannot metabolize the drug, causing severe bone marrow suppression. Use chlorambucil or cyclosporine in cats.
Phytonadione	**Vitamin K_1** — treats **anticoagulant rodenticide toxicosis**.
Terbutaline	**Beta-2 adrenergic agonist** — bronchodilation and airway relaxation. **Feline asthma.**
Metoclopramide	Crosses the BBB and **antagonizes dopamine in the CRTZ** to inhibit emesis; also a prokinetic.
Toceranib (Palladia)	Receptor tyrosine kinase inhibitor (c-KIT) — **canine mast cell tumors**.
Mitotane	Adrenocorticolytic — treats Cushing's, and can **cause iatrogenic Addison's**.
Amitraz	Alpha-2 agonist acaricide. **CONTRAINDICATED IN HORSES** — causes fatal impaction colic.

RAPID-FIRE PHARMACOLOGY RECALL

Ototoxic + nephrotoxic + NM blockade → **aminoglycosides**.
KCS in a dog on antibiotics → **sulfonamides/TMS**.
Doberman + sulfa → **type III hypersensitivity**.
Renal patient needing a tetracycline → **doxycycline**.
IV doxycycline in a horse → **collapse and death**.
Erythromycin in a horse → **fatal colitis**; in the mare of a treated foal → ***C. difficile***.
No anaerobic coverage → **fluoroquinolones and aminoglycosides**.
Best anaerobic coverage → **penicillins, clindamycin, metronidazole**.
NMDA antagonist → **ketamine**. Dopamine antagonist sedative → **acepromazine**.
Ryanodine receptor antagonist → **dantrolene** → **malignant hyperthermia in pigs**.
Ceiling effect → **buprenorphine and butorphanol**.
Azathioprine in a cat → **no TPMT** → **marrow suppression**.

Appendix A | Zoonoses & Public Health

Agent	Source	Human Disease & Prevention
Ancylostoma (hookworm)	Dogs, contaminated soil	**Cutaneous larval migrans** — itchy serpiginous skin tracks. Prevent with routine deworming and picking up feces.
Toxocara canis	Dogs, especially puppies	**Visceral and ocular larval migrans.** A leading infectious cause of childhood blindness. Deworm puppies from 2 weeks.
Baylisascaris	Raccoons	Devastating **neural** larva migrans in children.
Echinococcus granulosus	**Dog is the definitive host**	**Hydatid cyst disease** — cysts in liver and lungs; potentially fatal. Do not feed raw offal to dogs.
Toxoplasma gondii	**Cat is the definitive host**	**Congenital infection → chorioretinitis and mental retardation.** Oocysts need **1–5 days to sporulate**, so **daily litter box cleaning prevents transmission**. Pregnant women should avoid the litter box, garden with gloves, and not eat undercooked meat — which is actually the more common route.
Cryptosporidium	Many species; water	**Resistant to chlorination and standard disinfectants** → waterborne outbreaks. Severe in immunocompromised people.
Giardia	Many species; water	Diarrheal illness.
Gnathostoma	**Undercooked fish**	Migratory cutaneous swellings; rarely CNS disease.
Bartonella henselae	**Cats, via FLEAS**	**Cat scratch fever** — regional lymphadenopathy. Flea control is the prevention. Serious in immunocompromised people (bacillary angiomatosis).
Francisella tularensis	Cats/dogs hunting rabbits & rodents	**Tularemia.** Very low infectious dose; **select agent / potential biological weapon.** Warn the laboratory on any suspect submission.
Sporothrix schenckii	**Cats** — huge organism burden	**Sporotrichosis.** Transmits through **intact skin**. Wear gloves handling suspect feline nasal/cutaneous lesions.
Microsporum canis / ***T. mentagrophytes***	Cats; rodents/hamsters	**Ringworm.** *T. mentagrophytes* is the one spread from hamsters, and it does not fluoresce.
Sarcoptes and ***Cheyletiella***	Dogs, cats, rabbits	Papular, itchy dermatitis in owners — self-limiting once the animal is treated.
Leishmania	Dogs; sand fly	Visceral leishmaniasis.
Trypanosoma cruzi	Kissing bug feces	**Chagas disease** — cardiomyopathy.
Borrelia burgdorferi / ***Rickettsia rickettsii***	Shared tick exposure	**Lyme** (erythema migrans rash) and **RMSF**. The **dog is a sentinel**, not a direct source — people acquire both from the same ticks.
Chlamydia psittaci, ***Salmonella***, **rabies**	Various	Always on the differential list in a public-health question.

Appendix B | Breed Predispositions & Numbers to Memorize

B.1 Breed and Signalment Associations

Breed	Association
Doberman Pinscher	**Type III hypersensitivity to SULFA drugs.** Also von Willebrand disease, DCM, PLN.
Basset Hound	**Canine thrombopathia** — platelets fail to aggregate or degranulate.
Australian Shepherd	**Pelger-Huët anomaly** — benign hyposegmented neutrophils mimicking a severe left shift.
Color-dilute (smoke blue) Persian cat	**Chédiak-Higashi syndrome** — giant leukocyte granules, bleeding tendency.
Himalayan and Persian cats	**Basal cell tumors.**
Abyssinian, British Shorthair, Devon Rex, Cornish Rex	**Type B blood** — high transfusion reaction risk.
Soft-Coated Wheaten Terrier	**Protein-losing nephropathy** (and enteropathy) — distal limb edema and weight loss.
Greyhound and Pit Bull	**Babesiosis.** Pit Bulls specifically for *B. gibsoni* via bite transmission.
German Shepherd (dolichocephalic)	**Nasal aspergillosis.** Also severe chronic ehrlichiosis.
Beagle, Keeshond, Dachshund, Labrador, Golden Retriever, Vizsla	**Idiopathic epilepsy**, onset 1–5 years.
Cavalier King Charles Spaniel	**Syringomyelia** (and myxomatous mitral valve disease).
Labrador and Golden Retriever	**Lyme nephritis** — the protein-losing form of Lyme disease.

B.2 Numbers Worth Memorizing Cold

Number	Meaning
1 mL/kg pRBCs = 1% PCV	Transfusion dosing.
PCV < 20% with acute blood loss	Transfusion trigger.
SpO_2 90% = PaO_2 60 mmHg	The oxygen supplementation threshold.
$PaCO_2$ 35–45 mmHg	Normal ventilation. >45 hypoventilation, <30 hyperventilation.
PaO_2 < 50 mmHg	Threshold for visible cyanosis.
HCO_3^- 22–24 mEq/L	Normal bicarbonate.
Lactate < 2.5 mmol/L	Normal.
0.3–0.4 × BW × base deficit	Bicarbonate replacement formula.
TBW 60% · ICF 40% · ECF 20%	Body water. ECF is 75% interstitial, 25% intravascular.
Abdominal fluid : serum creatinine ≥ 2:1	Uroabdomen.
Blood glucose 20 mg/dL > abdominal fluid glucose	Septic peritonitis.
Glucose nadir 80–150 mg/dL	Insulin curve target, dogs and cats.
Feline mammary: >3 cm → 4–6 mo; <2 cm → 3 yr	Prognosis by size.
Canine mammary: ~50% benign	The 50/50 rule.
Vaccine sarcoma 3-2-1	3 months present · 2 cm · growing at 1 month.
Toxoplasma oocysts sporulate in 1–5 days	Why daily litter box cleaning works.
Foramen ovale closes within 48 hours	Neonatal circulation.

Number	Meaning
Dog 42 teeth · Cat 30 teeth	Dentition (pig 44, ruminant 32, horse 36–44).
Dog gestation 62–65 d · Cat 65 d	Estrus: dog 9 d, cat 7 d.
Marek's <16 wk vs. lymphoid leukosis >16 wk	Carried over from the avian guide — same exam, same day.

Appendix C | Topics to Study Elsewhere

READ THIS BEFORE YOU BUILD A STUDY SCHEDULE

The chapters in this guide go deep, but they do not cover everything. The topics below are **guaranteed NAVLE material that falls outside this guide's scope** — some appear only in passing, some not at all. **A guide that feels complete is not the same as coverage that is complete.** Study these from another resource.

Topic	Coverage Here	Why It Matters
Heartworm disease (*Dirofilaria immitis*)	**One passing mention** — only that ivermectin is "good for heartworm."	A guaranteed NAVLE topic. Know: life cycle and the 6-month prepatent period, antigen vs. microfilaria testing, caval syndrome, the **melarsomine 3-dose protocol**, strict exercise restriction, doxycycline for *Wolbachia*, and why you never give a fast-kill adulticide without preparation. Feline heartworm (HARD) is a different disease entirely.
Canine and feline endocrinopathies	**Nearly absent** — insulin curves and a passing mitotane reference only.	Missing entirely: **hyperadrenocorticism** (ACTH stim vs. LDDS vs. HDDS, PDH vs. adrenal tumor, trilostane), **hypoadrenocorticism** (Na:K ratio, the ACTH stim as the definitive test, crisis management), **hypothyroidism** in dogs, and **hyperthyroidism** in cats (methimazole, I-131, unmasking of renal disease). These are among the most heavily tested small animal topics.
FIV and FIP	**Absent.** Only the Rivalta test appears, and without naming FIP.	FIV transmission and testing; FIP pathogenesis (coronavirus mutation), effusive vs. non-effusive forms, and the diagnostic approach.
Canine parvovirus	**Absent** for dogs — parvovirus appears only for pigs, mink, and ferrets.	Signalment, crypt necrosis, diagnosis, treatment, vaccination timing and maternal antibody interference.
Canine distemper (dogs)	Mentioned only as a ferret disease and a type IV example.	Full clinical syndrome, hyperkeratosis, neurologic sequelae, vaccination.
Infectious tracheobronchitis	"Kennel cough" appears once, only as a test-statistics example.	*Bordetella*, canine parainfluenza, canine influenza; vaccination routes.
Leptospirosis	Mentioned only in a bovine differential.	**A major canine zoonosis** — serovars, acute kidney and liver injury, MAT interpretation, doxycycline, and vaccination.
Rabies	Appears in large animal contexts only.	Post-exposure protocols, quarantine periods, reporting duties. A near-certain exam item and a legal/public health obligation.
IVDD and orthopedics	**Absent.** Localization is covered but not the diseases.	Hansen type I vs. II, modified Frankel grading, the significance of deep pain, surgical timing. Plus **cranial cruciate rupture**, **hip dysplasia**, **patellar luxation**, **OCD**, and **panosteitis**.
GDV	One mention, only as an example of obstructive shock.	Signalment, "double bubble" radiograph, decompression, derotation, gastropexy, reperfusion arrhythmias.
Pyometra	One mention as an estradiol consequence.	Open vs. closed, cystic endometrial hyperplasia, the sick post-estrus bitch, OHE as treatment.
Brachycephalic obstructive airway syndrome	**Absent.**	Stenotic nares, elongated soft palate, everted saccules, hypoplastic trachea; anesthetic risk.
Dentistry	Tooth counts only.	Periodontal disease staging, feline tooth resorption, extraction indications.
Nutrition	Thiamine and taurine only in passing.	Life-stage feeding, obesity management, therapeutic diets, taurine-responsive DCM.

Topic	Coverage Here	Why It Matters
Behavior	**Absent.**	Separation anxiety, aggression assessment, house-soiling, feline inappropriate elimination.

WHERE THIS GUIDE IS STRONG

The chapters above go **unusually deep** on toxicology, clinical pathology and hematology, acid–base and critical care, vector-borne disease, systemic mycoses, and parasitology — several at a depth beyond what the NAVLE requires. The efficient plan is to **hold that ground with this guide** and spend new study time on the list above.